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KING ABDULAZIZ UNIVERSITY

FCIT
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كلية الحاسبات وتقنية المعلومات
جامعة الملك عبدالعزيز

# CPCS-381

# Human-Computer Interaction

# Project

## Flavor Code

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# Part 1 – Problem Definition and Context

## 1.1 Introduction

Flavor Code is an interactive cooking and recipe web application designed to help users discover, organize, and prepare meals in a more efficient and engaging way. The website aims to solve common problems users face while cooking, such as difficulty following recipe instructions, lack of dietary filtering options, and the inconvenience of constantly switching attention between cooking tasks and long recipe pages.

Many existing recipe platforms present information in a static and text-heavy format, which increases user frustration, especially for beginner cooks or users cooking under time pressure. In addition, users who follow specific dietary goals such as high-protein, healthy, vegan, or low-carb diets often struggle to quickly find suitable recipes that match their preferences.

## 1.2 Problem Importance and Impact

This problem is worth solving because cooking is a daily activity for many people, and poor recipe usability can negatively affect the overall cooking experience. Users may become overwhelmed, distracted, or discouraged from cooking at home due to unclear instructions and inefficient navigation.

A successful design can significantly improve the user experience by:

- Reducing stress during cooking.
- Helping users prepare meals more efficiently.
- Encouraging healthier eating habits.
- Supporting beginners with step-by-step guidance.
- Making recipe exploration more interactive and enjoyable.

By improving usability and accessibility, Flavor Code can encourage users to cook more confidently and rely less on unhealthy fast-food alternatives.

## 1.3 Proposed Solution

Flavor Code provides an interactive and user-centered solution through several integrated features that simplify the cooking process and improve usability. The website allows users to:

- Browse and search recipes easily.

- Filter recipes based on dietary categories such as healthy, high-protein, vegan, dessert, and low-carb meals.
- View detailed nutritional information for each recipe.
- Save favorite recipes to a personal profile.
- Access external cooking tutorials and nutrition resources.
- Use an interactive “Cook Mode” that presents recipes step-by-step with large readable text, cooking timers, progress tracking, and ingredient checklists.

The Cook Mode is specifically designed to reduce cognitive overload by showing one instruction at a time, allowing users to focus on cooking without unnecessary distractions.

## 1.4 Target Users and Context of Use

The primary target users of Flavor Code are:

- Beginner home cooks.
- Health-conscious individuals.
- Busy users looking for quick and simple meal preparation.

The website is mainly intended for users between the ages of 18–35 who regularly use smartphones, tablets, or laptops for cooking assistance.

**Context of Use**

Users interact with Flavor Code in different environments, including:

1. At home while preparing meals.
2. In kitchens where users may have wet or occupied hands.
3. During busy daily routines when quick access to recipes is important.

The website is commonly used:

- Before cooking to search for recipes.
- During cooking to follow step-by-step instructions.
- After cooking to save or revisit favorite meals.

The interface is designed to support users in real cooking environments by prioritizing readability, simplicity, large interaction areas, and minimal distractions.

## 1.5 Review of Similar Designs

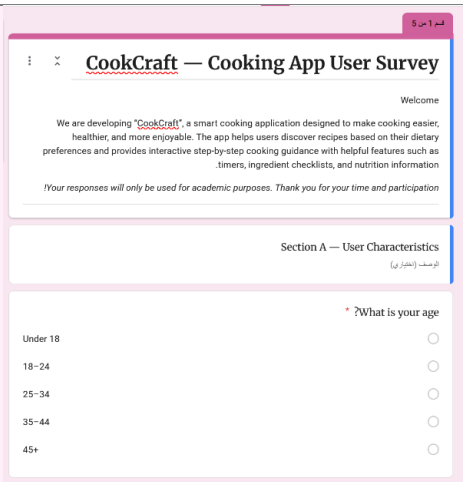
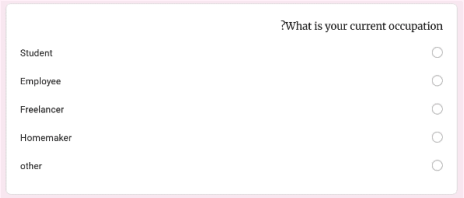
| WEBSITE | STRENGTHS | WEAKNESSES |
|---|---|---|
| **TASTY** | Provides clear recipe categories, visually organized recipe pages, detailed cooking instructions, and strong visual content through videos and images. The interface is simple and easy to follow. | The interface is not highly interactive and mainly focuses on displaying recipes and videos. Some recipes lack detailed nutritional information and advanced cooking support features, which can make the experience feel repetitive over time. |
| **SIDECHEF** | Offers a wide range of recipe categories, ingredient images, built-in timers, user reviews, nutritional information, and detailed step-by-step cooking guidance that supports users during meal preparation. | The interface contains many features and options, which can make navigation feel complex or overwhelming during cooking. Some users may find the interaction flow too detailed for quick recipe access. |
| **ALLRECIPES** | Provides detailed recipes, nutritional values, serving adjustment tools, ratings, reviews, and strong community interaction. Users can easily modify serving sizes based on their needs. | The website interface can appear crowded due to the large number of categories and content sections. First-time users may find navigation confusing because information is distributed across multiple areas. |
| **FLAVOR CODE** | Provides a clean and modern interface focused on simplicity and usability. Features include interactive serving adjustment, automatic ingredient scaling, nutritional information, and structured cooking instructions that reduce cognitive load. | Since Flavor Code is currently a prototype, some advanced community and personalization features are still limited compared to large commercial cooking platforms. |

The review of similar designs helped identify several successful usability practices used in modern recipe platforms. Tasty provides strong visual organization and clear recipe categories, while SideChef focuses heavily on interactive cooking support through guided instructions and cooking tools. Allrecipes offers useful community-based features such as ratings, reviews, and serving adjustments. Although these platforms provide useful features, many existing platforms still suffer from crowded interfaces, limited interactivity, or overly complex navigation. Based on this analysis, Flavor Code was designed to combine simplicity, readability, nutritional guidance, and interactive cooking support while reducing distraction and cognitive load for users during cooking tasks.

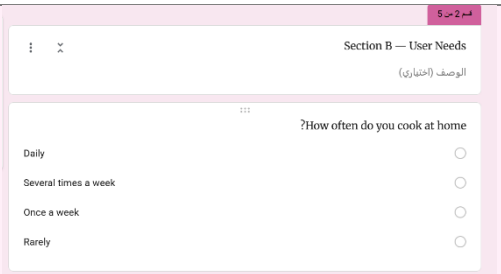
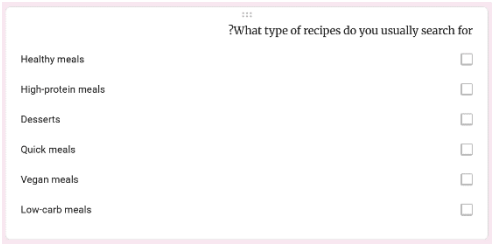
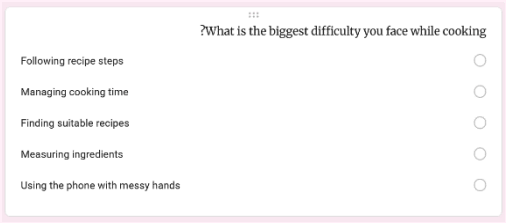
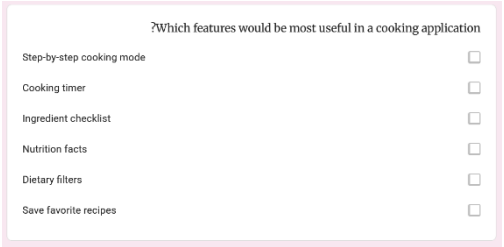
# Part 2 – Requirements

## 2.1 Questionnaire Draft

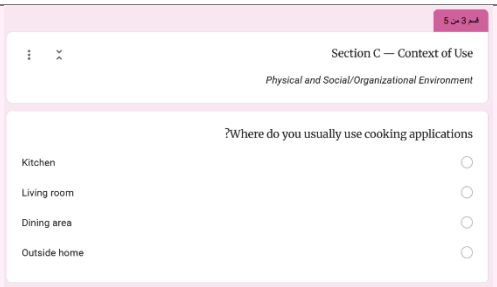
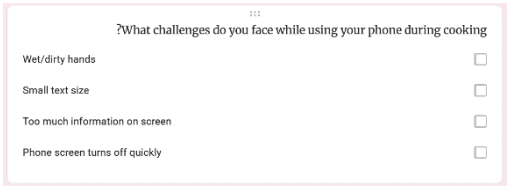
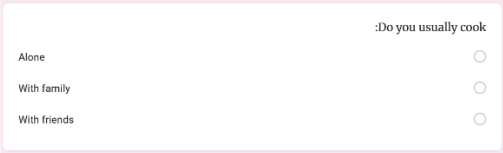
### 2.1.1 User Characteristics

| QUESTION | OPTIONS | PICTURE |
|---|---|---|
| **AGE** | • Under 18<br>• 18-24<br>• 25-34<br>• 35-44<br>• 45+ |  |
| **EDUCATION** | • Hugh school<br>• Diploma<br>• Bachelor's degree<br>• Master's degree<br>• Other |  |
| **JOB** | • Student<br>• Employee<br>• Freelancer<br>• Homemaker<br>• Other |  |
| **EXPERIENCE IN USING TECHNOLOGY** | • Beginner<br>• Intermediate<br>• Advanced |  |

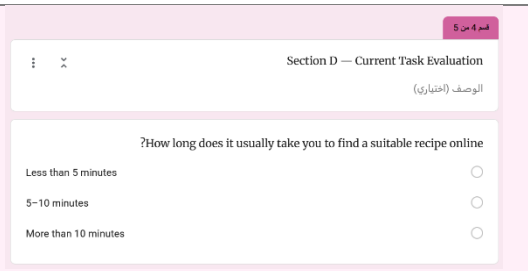
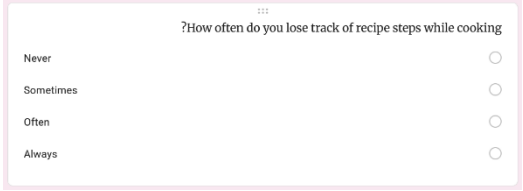
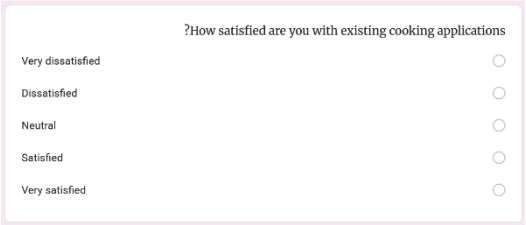
## 2.1.2 User Needs

| QUESTION | OPTIONS | PICTURE |
|---|---|---|
| **HOW OFTEN DO YOU COOK AT HOME?** | • Daily<br>• Several Times a week<br>• Once a week<br>• rarely |  |
| **WHAT TYPE OF RECIPES DO YOU USUALY SEARCH FOR?** | ▪ Healthy meals<br>• High-protein meals<br>• Desserts<br>• Quick meals<br>• Vegan meals<br>• Low-carb meals |  |
| **WHAT IS THE BIGGEST DIFFICULTY YOU FACE WHILE COOKING?** | • Following recipe steps<br>• Managing cooking time<br>• Finding suitable recipes<br>• Measuring ingredients<br>• Using the phone with messy hands |  |
| **WHICH FEATURES WOULD BE MOST USEFUL IN A COOKING APPLICATION?** | • Step-by-step cooking mode<br>• Cooking timer<br>• Ingredient checklist<br>• Nutrition facts<br>• Dietary filters<br>• Save favorite recipes |  |

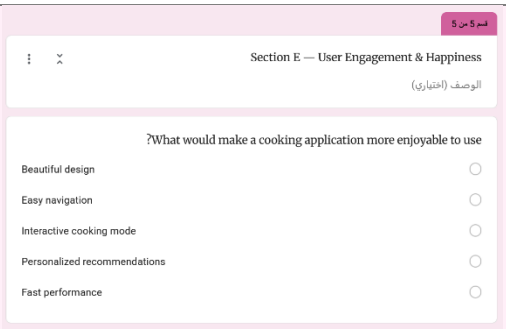
## 2.1.3 Context of Use

| QUESTION | OPTIONS | PICTURE |
|---|---|---|
| **WHERE DO YOU USUALLY USE COOKING APPLICATIONS?** | • Kitchen<br>• Living room<br>• Dining area<br>• Outside home |  |
| **WHAT CHALLENGES DO YOU FACE WHILE USING YOUR PHONE DURING COOKING?** | ▪ Wet/dirty hands<br>• Small text size<br>• Too much information on screen<br>• Phone screen turns off quickly |  |
| **DO YOU USUALLY COOK:** | • Alone<br>• With family<br>• With friends |  |
| **DO YOUR DAILY RESPONSIBILITIES AFFECT THE TIME YOU SPEND COOKING?** | • Yes<br>• No |  |

## 2.1.4 Current Task Evaluation

| QUESTION | OPTIONS | PICTURE |
|---|---|---|
| HOW LONG DOES IT USUALLY TAKE YOU TO FIND A SUITABLE RECIPE ONLINE? | • Less than 5 minutes<br>• 5–10 minutes<br>• More than 10 minutes |  |
| HOW OFTEN DO YOU LOSE TRACK OF RECIPE STEPS WHILE COOKING? | • Never<br>• Sometimes<br>• Often<br>• Always |  |
| HOW OFTEN DO YOU SWITCH BETWEEN APPS WHILE COOKING (TIMER, YOUTUBE, RECIPE APP)? | • Never<br>• Sometimes<br>• Often |  |
| HOW SATISFIED ARE YOU WITH EXISTING COOKING APPLICATIONS? | • Very dissatisfied<br>• Dissatisfied<br>• Neutral<br>• Satisfied<br>• Very satisfied |  |

## 2.1.5 User Engagement & Happiness

| QUESTION | OPTIONS | PICTURE |
|---|---|---|
| **WHAT WOULD MAKE A COOKING APPLICATION MORE ENJOYABLE TO USE?** | • Beautiful design<br>• Easy navigation<br>• Interactive cooking mode<br>• Personalized recommendations<br>• Fast performance |  |
| **WOULD YOU USE AN APPLICATION THAT PROVIDES STEP-BY-STEP COOKING GUIDANCE WITH TIMERS AND DIETARY FILTERS?** | • Yes<br>• Maybe<br>• No |  |

## 2.2 Tabulated Collected Data

We used Google spread sheets to tabulate collected data that consists of 30 responses.

| What is your age? | What is your education level? | What is your current occupation? | How would you rate your experience using technology and mobile application |
|---|---|---|---|
| 45+ | Bachelor's degree | Employee | Intermediate |
| 18–24 | Bachelor's degree | Student | Beginner |
| 18–24 | Bachelor's degree | Student | Advanced |
| 18–24 | High school | Freelancer | Advanced |
| 18–24 | Bachelor's degree | Student | Intermediate |
| 45+ | Diploma | other | Intermediate |
| 18–24 | Bachelor's degree | Employee | Advanced |
| 45+ | Bachelor's degree | Homemaker | Beginner |
| 25–34 | Bachelor's degree | Homemaker | Intermediate |
| 18–24 | Bachelor's degree | Student | Intermediate |
| 18–24 | High school | Student | Intermediate |
| 25–34 | Bachelor's degree | Freelancer | Advanced |
| Under 18 | Bachelor's degree | Student | Advanced |
| 18–24 | Bachelor's degree | Student | Advanced |
| 18–24 | Bachelor's degree | Student | Advanced |
| 18–24 | Diploma | Student | Advanced |
| 45+ | Bachelor's degree | Employee | Intermediate |
| 45+ | Bachelor's degree | Employee | Intermediate |
| 25–34 | Bachelor's degree | other | Advanced |
| 45+ | Bachelor's degree | Homemaker | Intermediate |
| 18–24 | Bachelor's degree | Student | Advanced |
| 18–24 | Bachelor's degree | other | Advanced |
| 18–24 | Bachelor's degree | Freelancer | Beginner |
| 45+ | High school | Homemaker | Beginner |
| Under 18 | Other | Student | Intermediate |
| 35–44 | Bachelor's degree | Employee | Beginner |

*Figure 1. Data Collected for User Characteristics.*

| How often do you cook at home? | What type of recipes do you usually search for? | What is the biggest difficulty you face while cooking? | Which features would be most useful in a cooking application? |
|---|---|---|---|
| Several times a week | Quick meals | Finding suitable recipes | Step-by-step cooking mode |
| Rarely | Healthy meals | Using the phone with messy hands | Step-by-step cooking mode |
| Several times a week | Healthy meals | Managing cooking time | Dietary filters |
| Several times a week | High-protein meals | Measuring ingredients | Nutrition facts |
| Several times a week | Healthy meals | Finding suitable recipes | Save favorite recipes |
| Rarely | Vegan meals | Using the phone with messy hands | Ingredient checklist |
| Several times a week | Desserts, Quick meals | Following recipe steps | Step-by-step cooking mode, Cooking timer |
| Daily | Healthy meals, High-protein meals, Desserts, Quick meals | Finding suitable recipes | Ingredient checklist, Nutrition facts, Save favorite recipes |
| Rarely | Quick meals | Following recipe steps | Save favorite recipes |
| Several times a week | Healthy meals | Using the phone with messy hands | Save favorite recipes |
| Several times a week | Healthy meals, High-protein meals, Quick meals | Measuring ingredients | Step-by-step cooking mode, Cooking timer, Ingredient checklist, Save favorite recipes |
| Several times a week | Healthy meals | Managing cooking time | Step-by-step cooking mode, Cooking timer, Ingredient checklist, Nutrition facts, Dietary filters, Save favorite recipes |
| Rarely | Healthy meals, High-protein meals, Desserts, Quick meals | Measuring ingredients | Step-by-step cooking mode, Ingredient checklist, Nutrition facts, Save favorite recipes |
| Once a week | Healthy meals, High-protein meals, Quick meals | Managing cooking time | Step-by-step cooking mode, Cooking timer |
| Several times a week | Healthy meals, Quick meals | Managing cooking time | Ingredient checklist |
| Several times a week | Quick meals | Finding suitable recipes | Step-by-step cooking mode, Cooking timer, Dietary filters, Save favorite recipes |
| Several times a week | High-protein meals, Desserts, Quick meals | Finding suitable recipes | Step-by-step cooking mode, Ingredient checklist |
| Daily | Healthy meals, High-protein meals, Quick meals, Vegan meals, Low-carb meals | Using the phone with messy hands | Step-by-step cooking mode, Cooking timer, Ingredient checklist, Nutrition facts, Dietary filters, Save favorite recipes |
| Once a week | High-protein meals, Quick meals | Finding suitable recipes | Ingredient checklist, Nutrition facts, Save favorite recipes |
| Once a week | Healthy meals, Desserts, Quick meals | Managing cooking time | Step-by-step cooking mode |
| Daily | Healthy meals, Desserts, Quick meals, Low-carb meals | Managing cooking time | Step-by-step cooking mode, Cooking timer |
| Rarely | Healthy meals, High-protein meals, Quick meals | Measuring ingredients | Step-by-step cooking mode |
| Daily | Healthy meals | Using the phone with messy hands | Step-by-step cooking mode |
| Daily | Low-carb meals | Finding suitable recipes | |
| Rarely | Healthy meals, High-protein meals, Quick meals | Finding suitable recipes | Step-by-step cooking mode, Ingredient checklist, Nutrition facts, Dietary filters, Save favorite recipes |
| Daily | Healthy meals, High-protein meals | Using the phone with messy hands | Ingredient checklist |
| Daily | Desserts | Finding suitable recipes | Ingredient checklist |
| Daily | Healthy meals, High-protein meals, Desserts, Low-carb meals | Following recipe steps | Step-by-step cooking mode, Save favorite recipes |
| Rarely | Desserts, Quick meals | Finding suitable recipes | Step-by-step cooking mode, Cooking timer |
| Rarely | Quick meals | Finding suitable recipes | Save favorite recipes |

*Figure 2. Collected Data for User Needs.*

| Where do you usually use cooking applications? | What challenges do you face while using your phone during cooking? | Do you usually cook: | Do your daily responsibilities affect the time you spend cooking? |
|---|---|---|---|
| Outside home | Phone screen turns off quickly | Alone | Yes |
| Living room | Phone screen turns off quickly | With friends | Yes |
| Kitchen | Phone screen turns off quickly | Alone | Yes |
| Dining area | Phone screen turns off quickly | Alone | No |
| Kitchen | Wet/dirty hands | Alone | Yes |
| Kitchen | Wet/dirty hands | With family | Yes |
| Kitchen | Wet/dirty hands, Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands, Small text size | Alone | No |
| Kitchen | Wet/dirty hands | Alone | Yes |
| Kitchen | Wet/dirty hands, Phone screen turns off quickly | Alone | No |
| Kitchen | Wet/dirty hands | With family | Yes |
| Living room | Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands | Alone | Yes |
| Kitchen | Too much information on screen, Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands, Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands, Small text size, Too much information on screen, Phone sc | Alone | No |
| Kitchen | Wet/dirty hands, Too much information on screen | Alone | No |
| Kitchen | Wet/dirty hands, Small text size, Too much information on screen, Phone sc | With family | No |
| Kitchen | Wet/dirty hands, Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands, Too much information on screen | Alone | Yes |
| Kitchen | Wet/dirty hands, Small text size, Too much information on screen, Phone sc | With family | Yes |
| Kitchen | Wet/dirty hands, Small text size | With friends | Yes |
| Kitchen | Wet/dirty hands | With family | Yes |
| Kitchen | Wet/dirty hands | Alone | Yes |
| Kitchen | Wet/dirty hands, Phone screen turns off quickly | Alone | Yes |
| Kitchen | Wet/dirty hands | With family | Yes |
| Living room | Small text size | With family | Yes |
| Kitchen | Wet/dirty hands | Alone | Yes |
| Kitchen | Wet/dirty hands | Alone | No |
| Kitchen | | With family | Yes |

*Figure3. Collected Data for Context of Use.*

| What would make a cooking application more enjoyable to use? | Would you use an application that provides step-by-step cooking guidance with timers and dietary filters? |
|---|---|
| Interactive cooking mode | yes |
| Interactive cooking mode | yes |
| Beautiful design | maybe |
| Interactive cooking mode | yes |
| Easy navigation | maybe |
| Beautiful design | yes |
| Interactive cooking mode | |
| Beautiful design | yes |
| Fast performance | yes |
| Fast performance | maybe |
| Easy navigation | yes |
| Personalized recommendations | yes |
| Easy navigation | no |
| Easy navigation | maybe |
| Personalized recommendations | maybe |
| Beautiful design | maybe |
| Personalized recommendations | maybe |
| Interactive cooking mode | yes |
| Personalized recommendations | yes |
| Beautiful design | maybe |
| Easy navigation | yes |
| Easy navigation | yes |
| Beautiful design | maybe |
| Easy navigation | no |
| Personalized recommendations | yes |
| Easy navigation | yes |
| Beautiful design | yes |
| Interactive cooking mode | maybe |
| Easy navigation | yes |
| Fast performance | maybe |

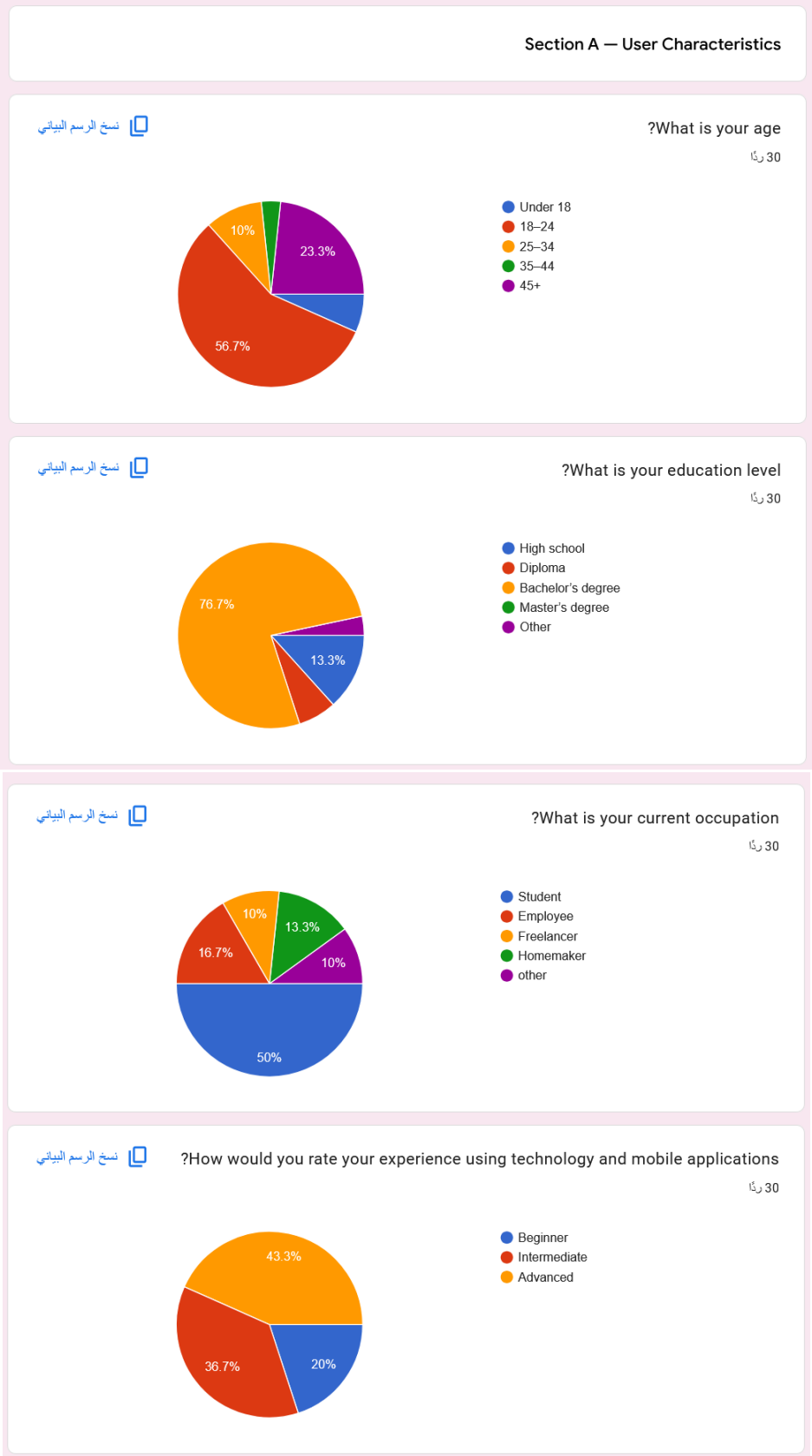
*Figure4. Collected data for Current Task Evaluation.*

| How long does it usually take you to find a suitable recipe online? | How often do you lose track of recipe steps while cooking? | How often do you switch between apps while cooking (timer, YouTube, recipe app)? | How satisfied are you with existing cooking applications? |
|---|---|---|---|
| More than 10 minutes | Sometimes | Sometimes | Very dissatisfied |
| Less than 5 minutes | Sometimes | Sometimes | Dissatisfied |
| More than 10 minutes | Sometimes | Never | Satisfied |
| More than 10 minutes | Sometimes | Often | Satisfied |
| 5–10 minutes | Sometimes | Sometimes | Satisfied |
| More than 10 minutes | Never | Sometimes | Neutral |
| 5–10 minutes | Sometimes | Never | Neutral |
| More than 10 minutes | Sometimes | Sometimes | Neutral |
| More than 10 minutes | Often | Often | Very dissatisfied |
| 5–10 minutes | Sometimes | Never | Very satisfied |
| More than 10 minutes | Sometimes | Often | Very satisfied |
| Less than 5 minutes | Sometimes | Never | Neutral |
| More than 10 minutes | Sometimes | Never | Neutral |
| 5–10 minutes | Sometimes | Sometimes | Neutral |
| 5–10 minutes | Often | Never | Neutral |
| 5–10 minutes | Sometimes | Sometimes | Dissatisfied |
| 5–10 minutes | Always | Never | Satisfied |
| 5–10 minutes | Often | Sometimes | Very satisfied |
| More than 10 minutes | Sometimes | Often | Neutral |
| 5–10 minutes | Always | Sometimes | Neutral |
| Less than 5 minutes | Often | Often | Neutral |
| 5–10 minutes | Sometimes | Sometimes | Satisfied |
| Less than 5 minutes | Sometimes | Sometimes | Neutral |
| Less than 5 minutes | Never | Never | Neutral |
| More than 10 minutes | Sometimes | Sometimes | Satisfied |
| 5–10 minutes | Often | Sometimes | Satisfied |
| 5–10 minutes | Sometimes | Sometimes | Very dissatisfied |
| More than 10 minutes | Sometimes | Sometimes | Neutral |
| More than 10 minutes | Always | Sometimes | Dissatisfied |
| 5–10 minutes | Sometimes | Sometimes | Neutral |

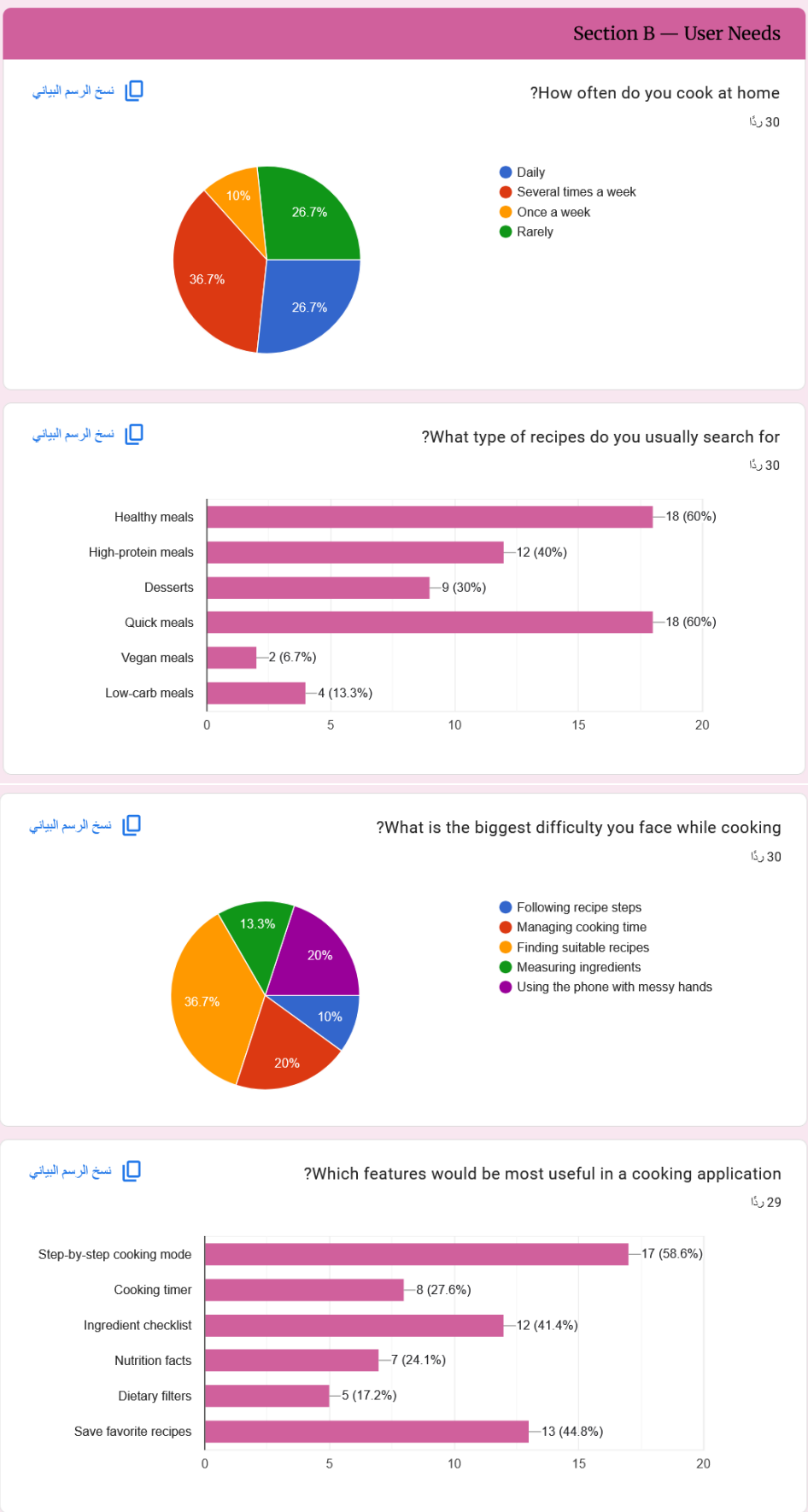
*Figure5. Collected Data for User Engagement & Happiness.*

# 2.3 Data Analysis

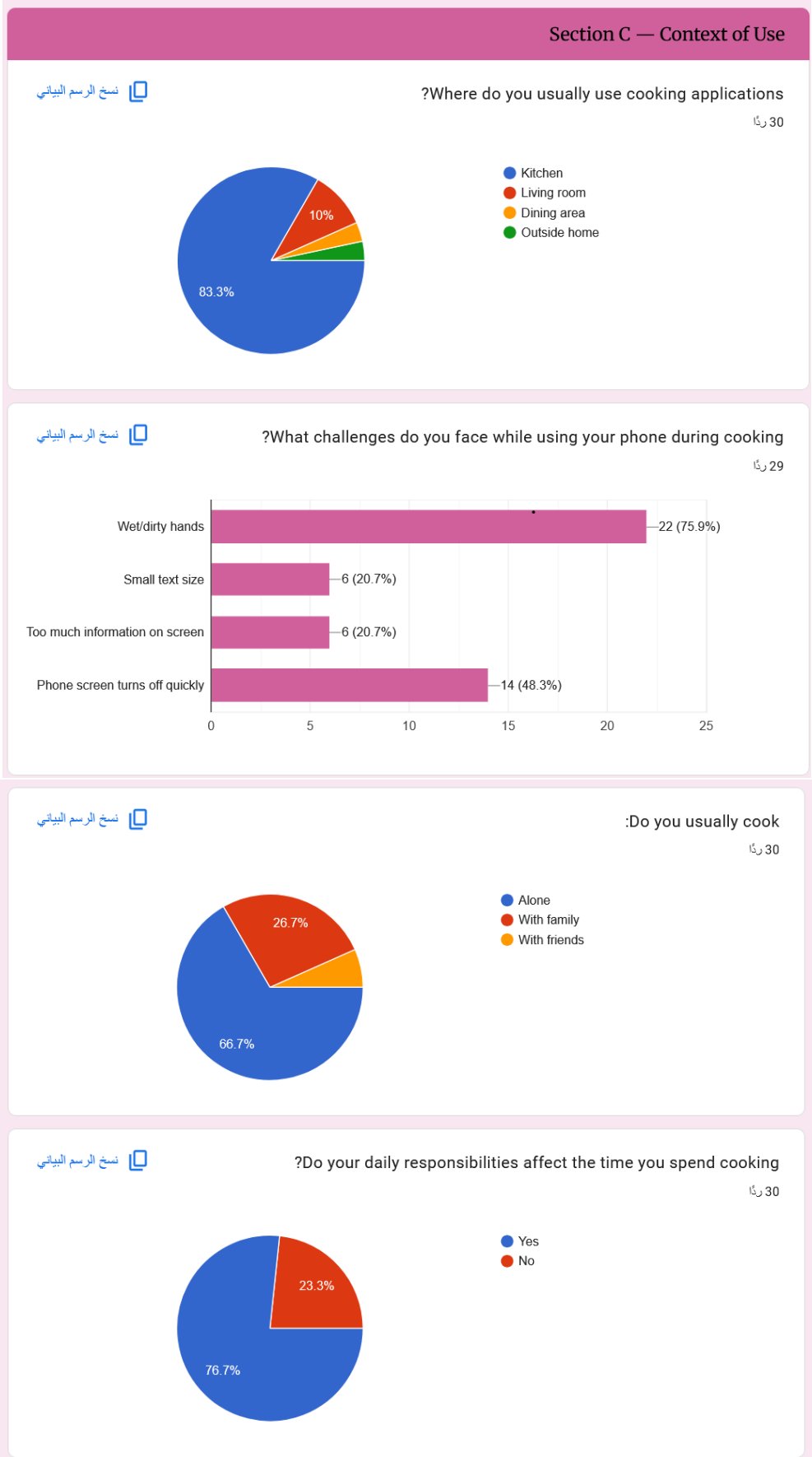
## 2.3.1 User Characteristics



## 2.3.2 User Needs



## 2.3.3 Context of Use

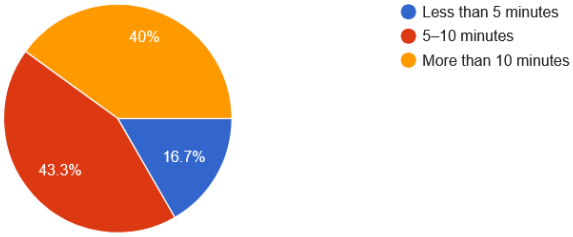


## 2.3.4 Current Task Evaluation

### Section D — Current Task Evaluation

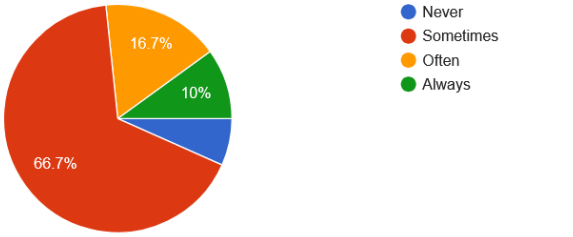
?How long does it usually take you to find a suitable recipe online

30 ردًا



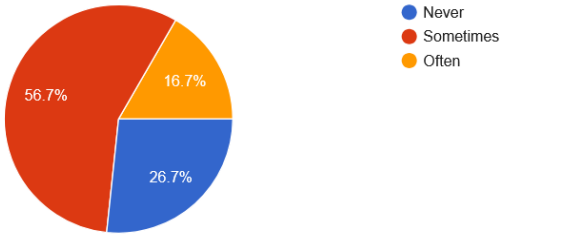

?How often do you lose track of recipe steps while cooking

30 ردًا



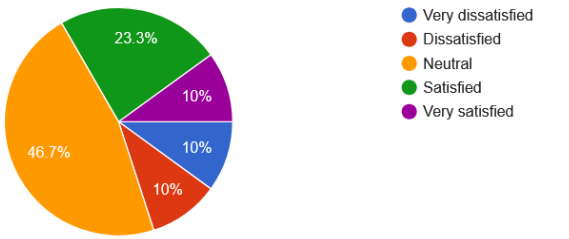

How often do you switch between apps while cooking (timer, YouTube, recipe ?app)

30 ردًا

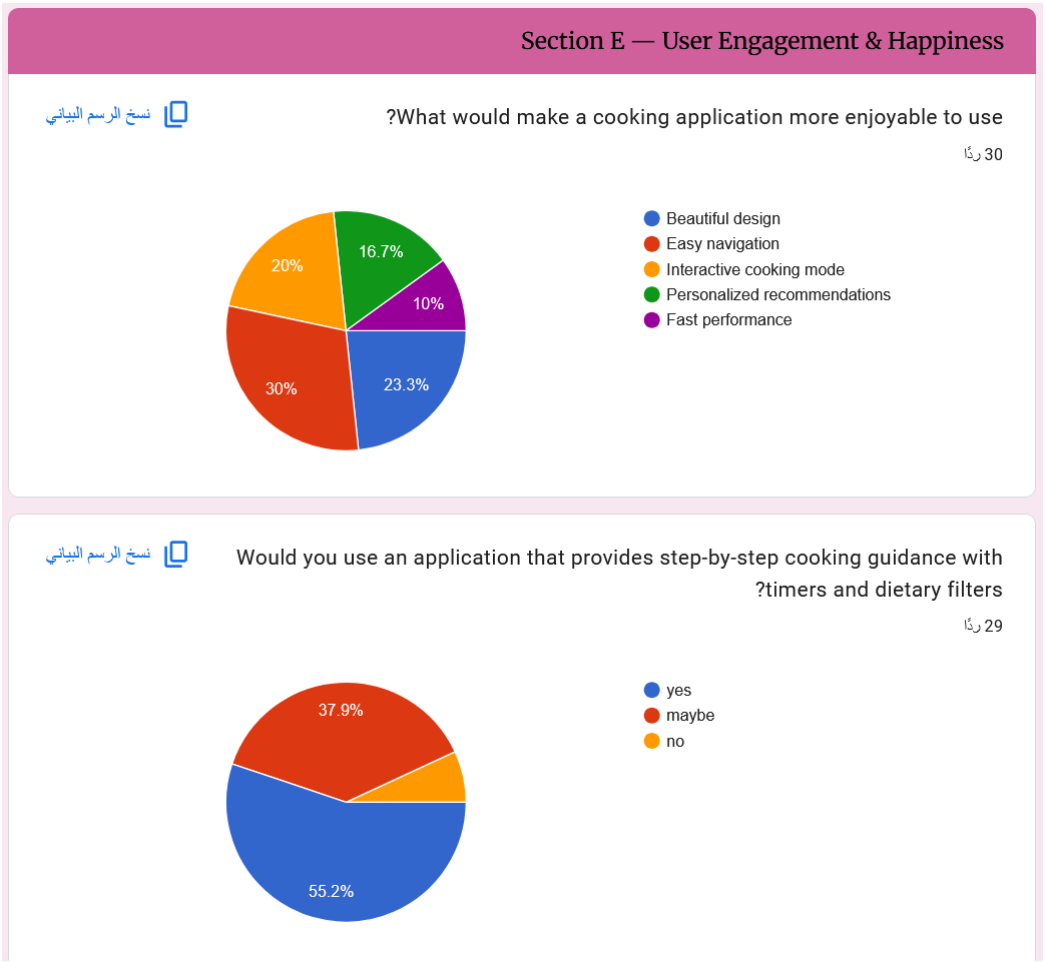



?How satisfied are you with existing cooking applications

30 ردًا



## 2.3.5 User Engagement & Happiness



## 2.4 Defined Interface Requirements

According to the collected data from the questionnaire responses, these are the website's requirements that has been obtained and must be applicable.

### 2.4.1 Category and Dietary Recipes' Filtering

- **Empirical Source of Data:** Based on the results in section *2.3.2* **60%** of users need a cooking recipe service that provides them with ***healthy meals*** and **40%** of users also require ***high-protein meals***.

- **Implementation in FlavorCode:** The web application provides horizontal filtering in the form of interactive "pill" filter triggers that allow users to search for recipes based on their category by displaying a collection of recipes that fall under the category that "pill" button has been clicked.

### 2.4.2 Isolated Step-by-step Instruction Delivery (Cook Mode)

- **Empirical Source of Data:** As shown in section 2.3.2**,** section 2.3.4 and section 2.3.5, some users find it somewhat difficult to follow a recipe step-by-step and would use a feature to make it easier.

- **Implementation in FlavorCode:** To reduce cognitive load and support sequential tracking, the web application incorporates full-screen interactive interface called **"Cook Mode"** that is activated and exited by user. The **"Cook Mode"** feature provides the following:

    - **Task Isolation:** Hides all background visual noise helping user concentrate on the current task one step at a time and displays the recipe's name.

    - **Linear Navigation:** Gives user the freedom to go **"Back"**, **"Next Step"** and **"Finish Cooking"** by interactive buttons visible on the screen.

    - **Enhanced Readability:** Utilizes large, bold and highly visible font to insure that instruction are visible from a distance like from the kitchen's counter top.

    - **Tips on Each Step:** The **"Cook Mode"** deliver specialized tips for every step to enhance user's cooking experience.

    - **On-Screen Cooking Tools:** Includes built-in timer, ingredient checklist and resources. The interface shows the expected time to complete each step with ability to **"Start"** and **"Pause"** timer. An ingredient checklist that is visible throughout the cooking process.

Resources of recipe that can be beneficial from a video that shows the steps to the resources of the dietary information.

- **Exit Button:** User can exit **"Cook Mode"** at any given time even if the steps has not been completed.

### 2.4.3 Profile Personalization (My Kitchen)

- **Empirical Source of Data:** Section 2.3.2 shows that **44.8%** of users think that having ***save favorite recipes*** is a useful feature.

- **Implementation in FlavorCode:** All recipe includes an interactive "Save Recipe" heart icon button that bookmark each recipe. Clicking the heart provides immediate user feedback via a toast notification **"Saved to My Kitchen"** and automatically saves the recipe into their private **"My Kitchen"** collection page. In addition to a profile interface that shows saved recipes, recent cooking history and dietary preferences.

### 2.4.4 Dynamic Nutrition Dashboard Tool

- **Empirical Source of Data:** Section 2.3.2 shows that individuals are very health-conscious care deeply about viewing accurate nutritional information on what they consume.

- **Implementation in FlavorCode:** FlavorCode resolves this requirement by having a dynamic ingredient side bar with clear nutritional information per serving. When user changes the number of servings using portion control, the website automatically shows the updated quantities of ingredients and nutritional information.

### 2.4.5 Error Prevention & Large Buttons

- **Empirical Source of Data:** From the questionnaire responses specifically in section 2.3.1, individuals experience frustration when cooking with messy/wet hand due to the small buttons on the screen.

- **Implementation in FlavorCode:** The website offers buttons that are large and clickable with rounded edges (i.e. **"Start"**, **"Next Step"** and **"Exit Cook Mode"** buttons). This makes them easier to click under kitchen stress.

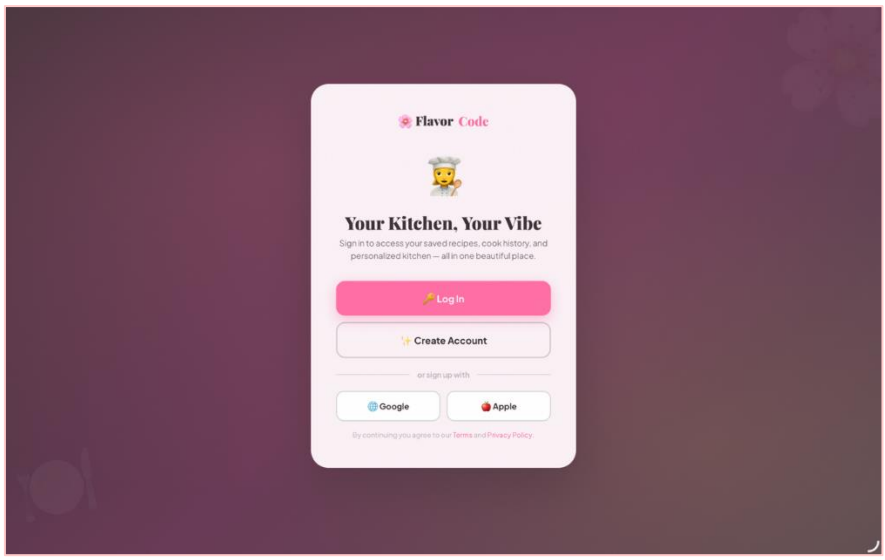
# Part 3 – Prototype

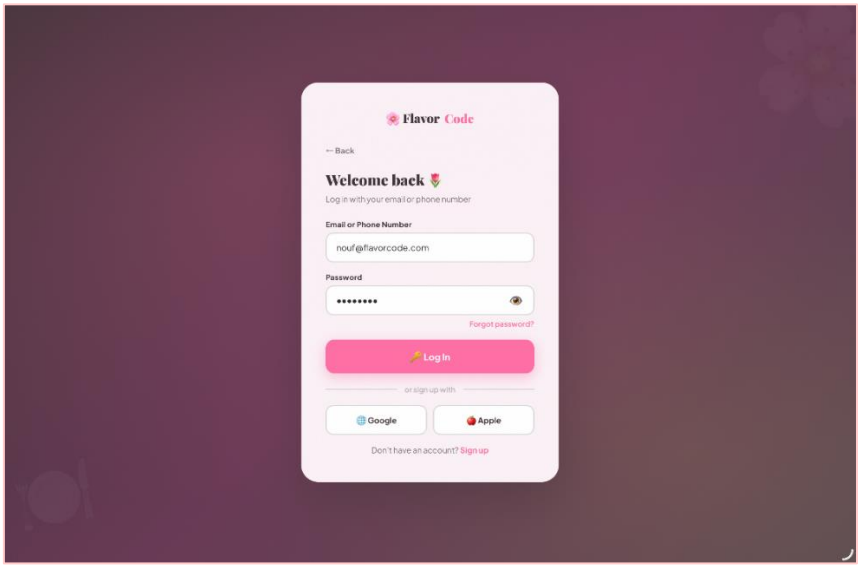
## 3.1 Prototype Development Tool

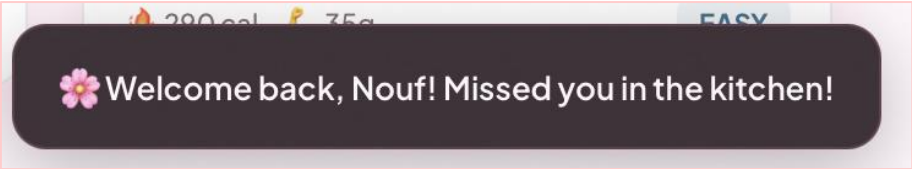
Our prototype was built using Claude (Anthropic) as the primary development tool. We provided detailed prompts describing the app's structure, features, and design vision — including the recipe browsing experience, cook mode, and authentication flow. Claude generated the website based on those descriptions. We then manually refined the output by adjusting colors, typography, layout properties, and specific UI behaviors to better match our design intent.

**Interface 1. Flavor Code Authentication**

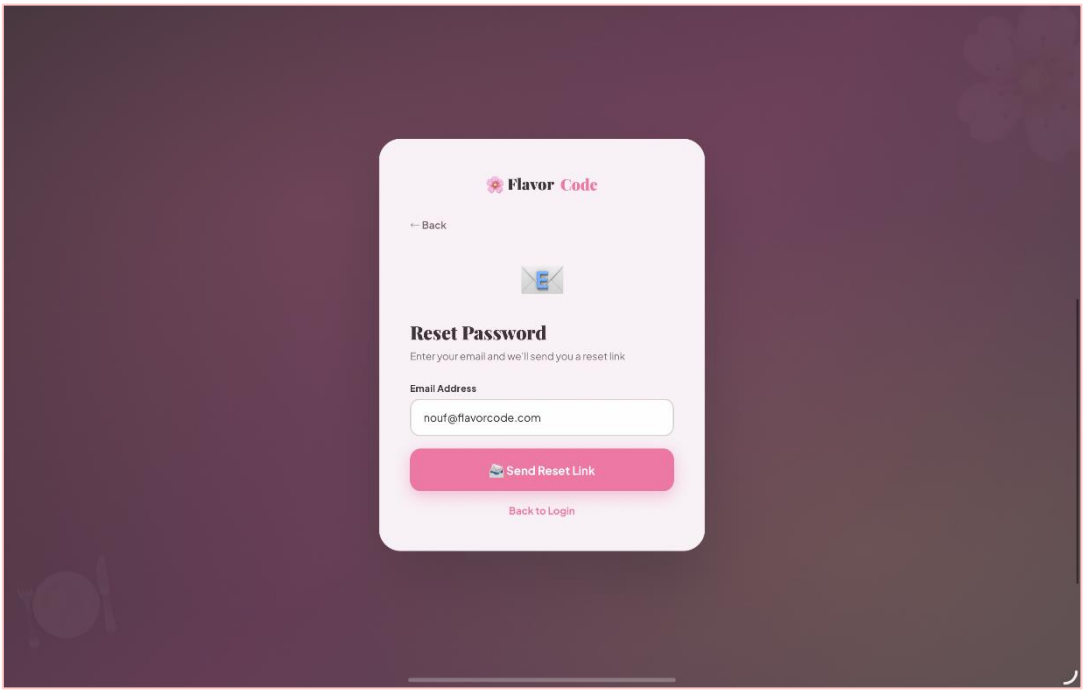
### A- log in

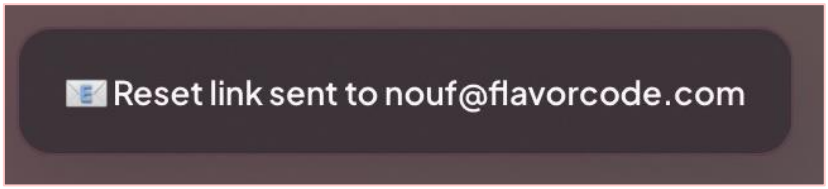


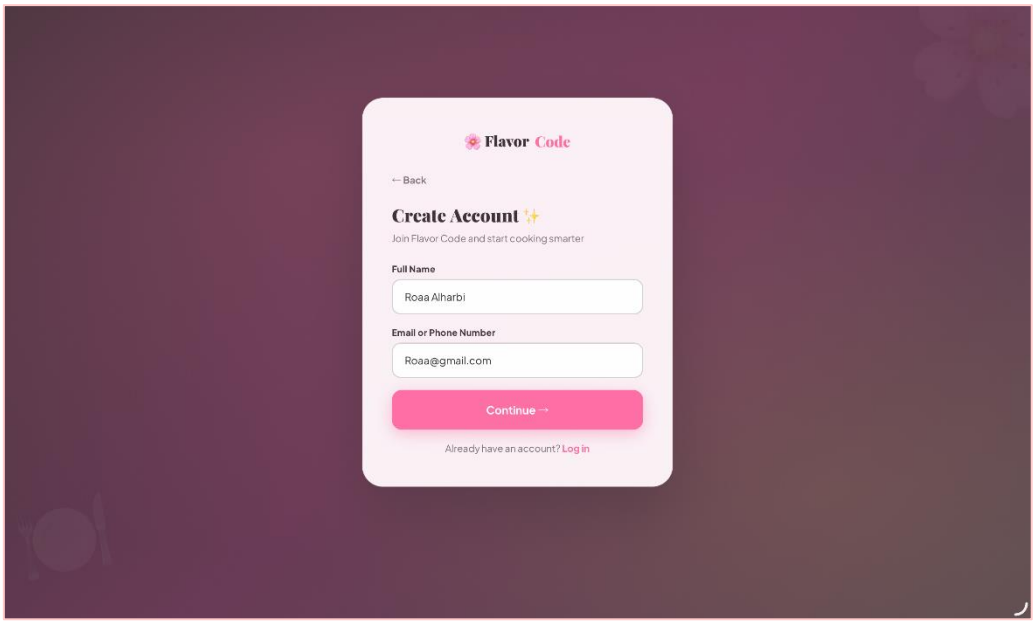


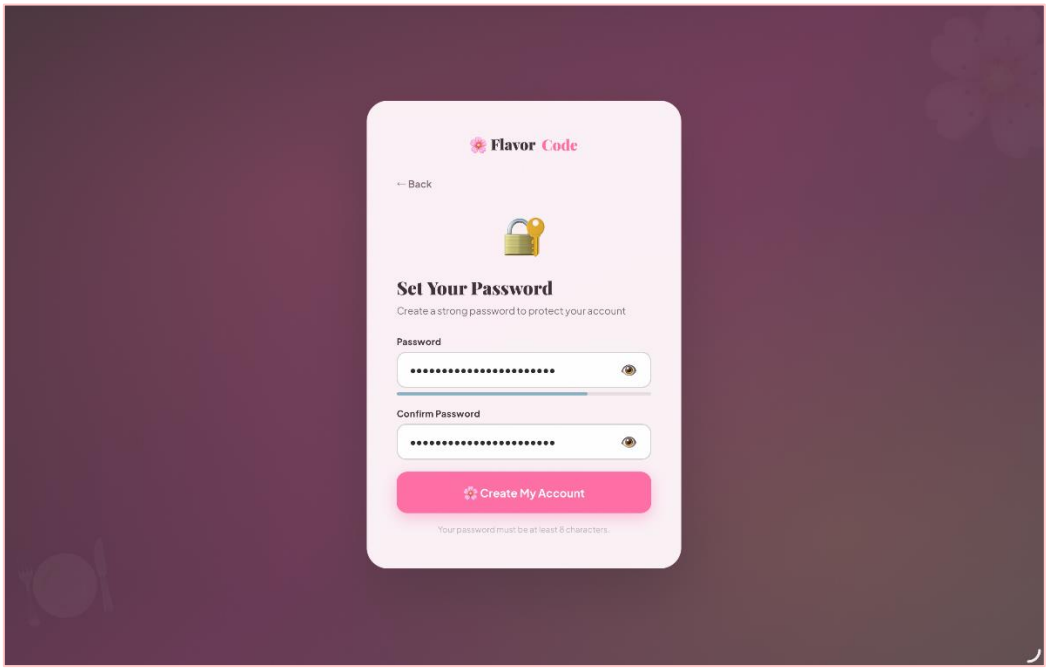

## B- Forget Password?

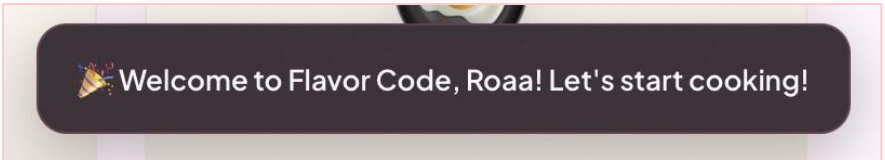


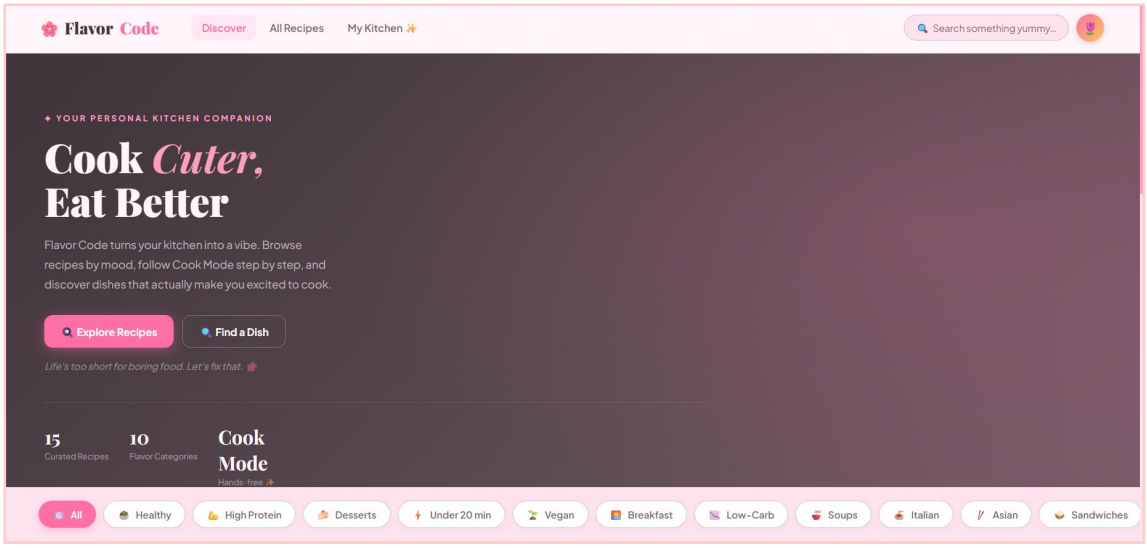
## C- Sign Up

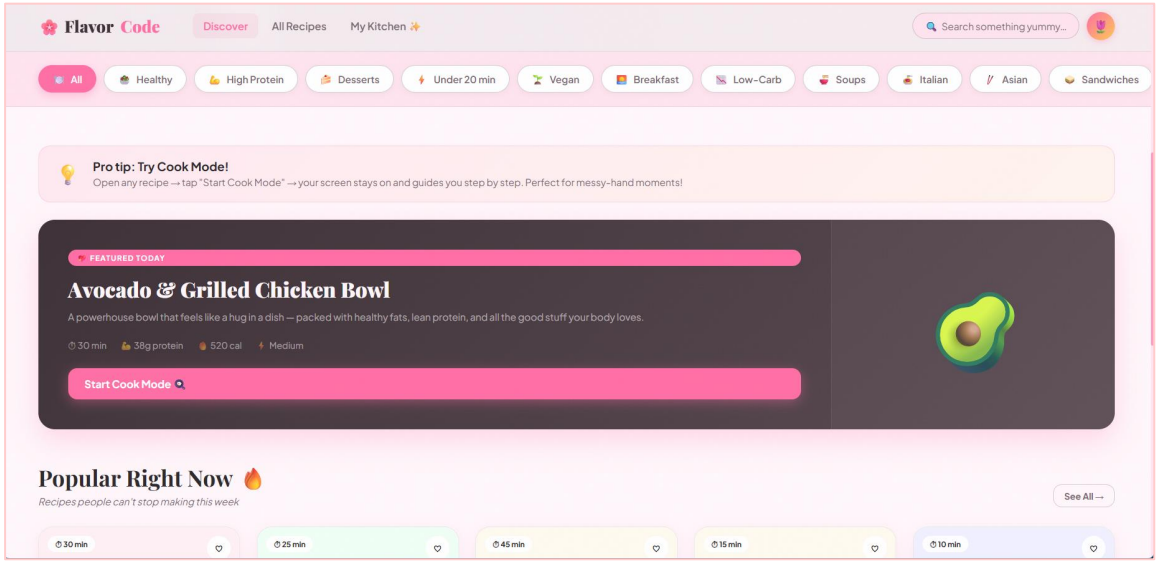

Flavor Code
← Back
Create Account ✨
Join Flavor Code and start cooking smarter
Full Name
Roaa Alharbi
Email or Phone Number
Roaa@gmail.com
Continue →
Already have an account? Log in

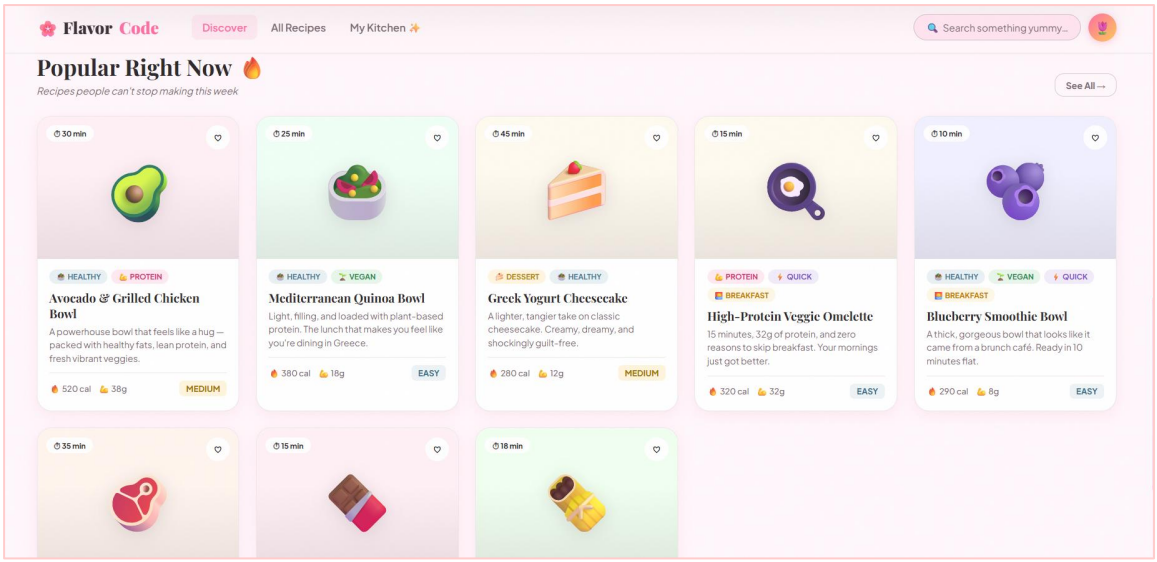

Flavor Code
← Back
Set Your Password
Create a strong password to protect your account
Password
Confirm Password
Create My Account
Your password must be at least 8 characters.


🎉 Welcome to Flavor Code, Roaa! Let's start cooking!

## 2. Discover Interface






**Discover Interface Description**

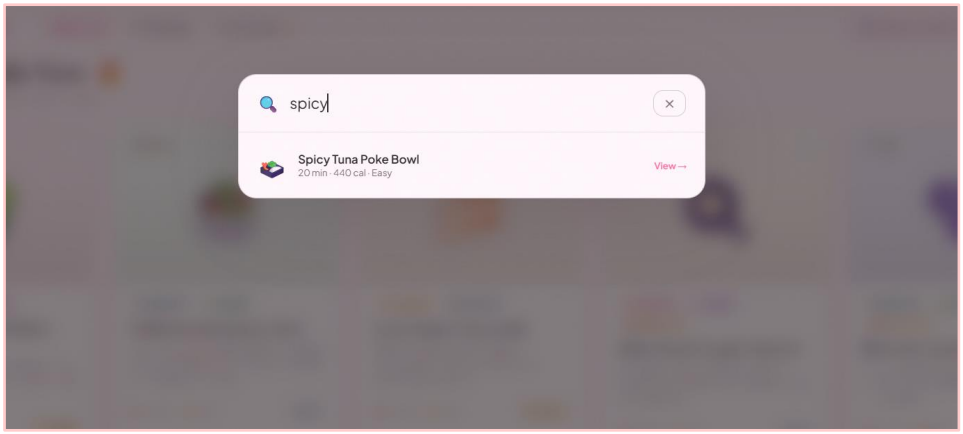
Hero Branding: The section features a bold, stylish headline “Cook Cuter, Eat Better” that defines the platform’s identity.

Content Overview: It displays high-level statistics, specifically highlighting “15 Curated Recipes” and “10 Flavor Categories” to show the variety available.

Interactive Search: A prominent, rounded search bar is positioned at the top with the inviting micro-copy “Search something yummy…”.

Category Filtering: A horizontal navigation bar allows users to filter recipes through stylized “pills” like Healthy, Desserts, Vegan, and Asian, each accompanied by a representative icon.

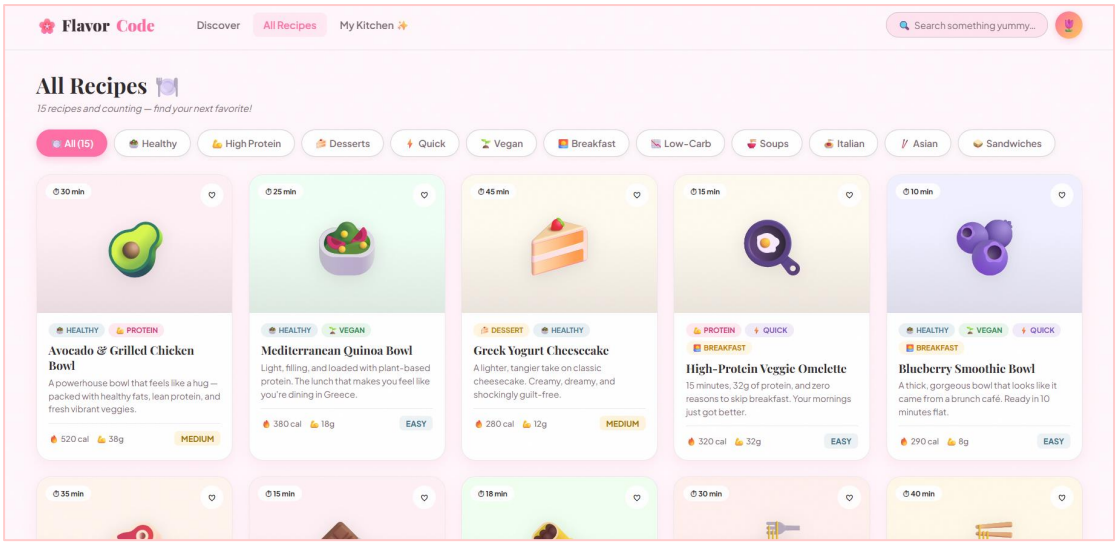
- *Popular Recipes Section:* A “Popular Right Now” header introduces a gallery of trending dishes, using a flame emoji to indicate popularity.
- *Recipe Card Layout:* Each recipe is presented in a clean, white card with rounded corners, featuring a central 3D food illustration (like an avocado or a steak).
- *Visual Information Density:* The cards provide immediate data at a glance, including preparation time (e.g., 30 min), calorie count, and difficulty labels (Easy/Medium).
- *Personalization Tools:* Every recipe card includes a heart icon in the top-right corner, allowing users to quickly save dishes to their favorites.

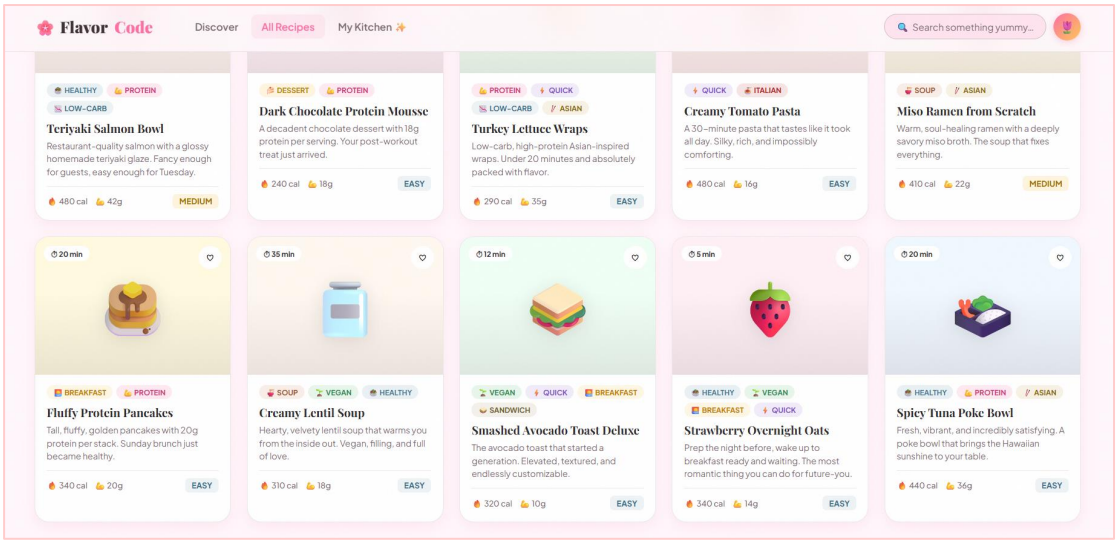



**Search Interface**

- *Overlay Search Bar:* A minimalist, focused pop-up that appears upon interaction to prioritize user search.
- *Live Results:* Displays real-time recipe matches as the user types (e.g., "spic" for Spicy Tuna Poke Bowl).
- *Quick Summary:* Shows immediate recipe data including prep time, calories, and difficulty level within the result.
- *One-Tap Access:* Includes a "View" shortcut for instant navigation to the recipe details.

### 3. All Recipes Interface






## All Recipes Interface Description

- *Comprehensive Recipe Gallery*: This section serves as the central hub of the platform, showcasing the full collection of "15 recipes and counting" in a visually organized grid.

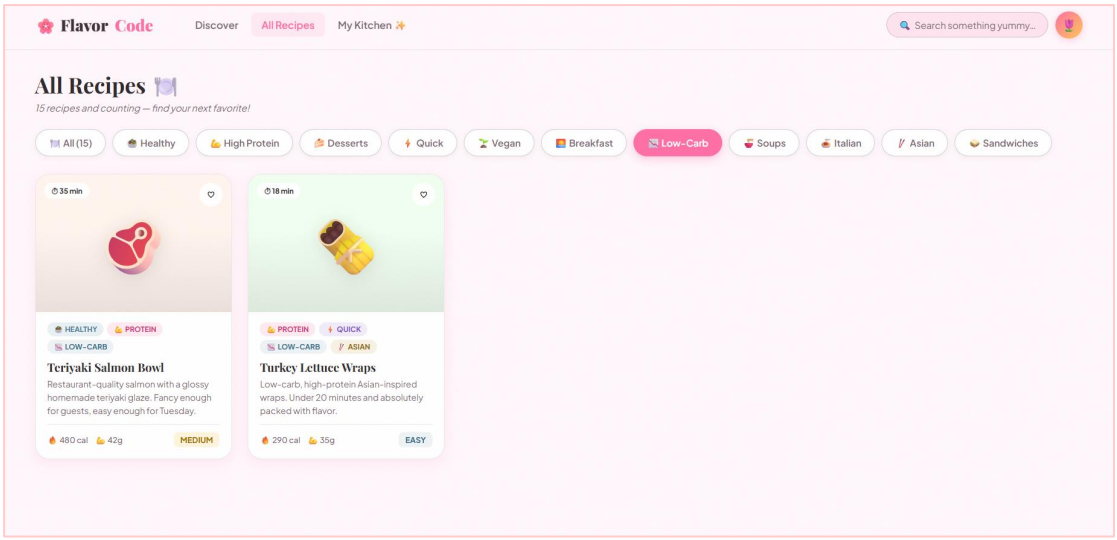
- *Advanced Filtering System:* The interface features an extensive horizontal navigation bar with interactive "pill" buttons, allowing users to sort the entire library by specific categories such as Healthy, High Protein, Desserts, Soups, and Sandwiches.

- *Consistent Visual Language*: The page maintains the site's aesthetic integrity by using uniform recipe cards, each highlighted by a high-quality 3D food illustration that makes browsing an engaging experience.

- *Quick-Glance Metadata:* Each card is strategically designed to provide essential data points—such as preparation time (ranging from 5 to 45 minutes) and a difficulty label (Easy/Medium)—helping users choose a meal that fits their schedule and skill level.

- *Nutritional and Dietary Tagging*: Every recipe includes clear dietary labels (e.g., Vegan, Quick, Breakfast) and transparent nutritional facts, specifically displaying calorie counts and protein grams.

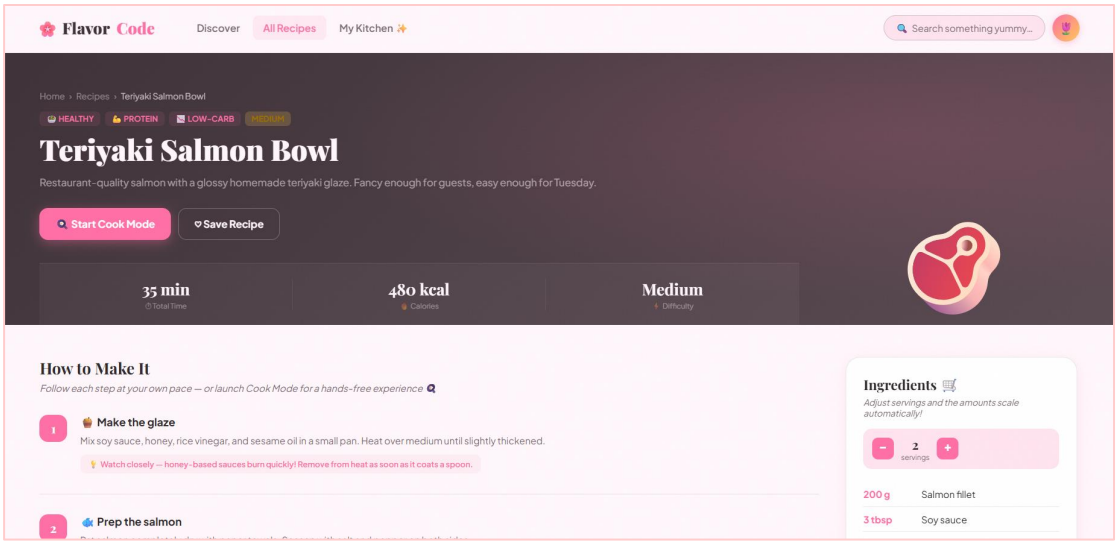
• *Seamless Navigation:* The "All Recipes" tab is clearly highlighted in the top navigation bar, ensuring users always know their current location within the site's architecture.

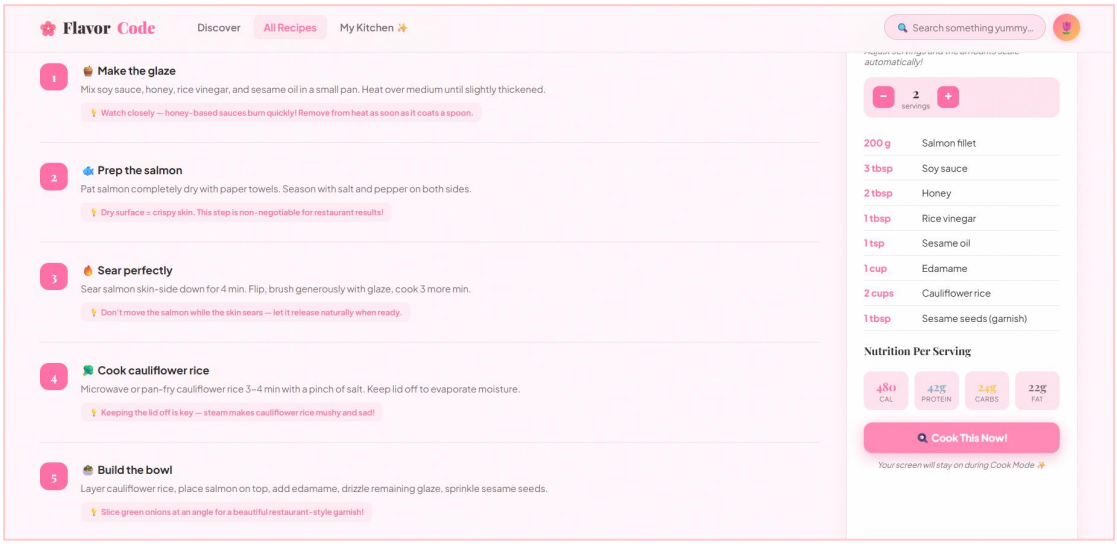



**Filtered View: Low-Carb Category**

- *Refined Discovery:* Dynamically updates the recipe grid to display only dishes that align with low-carbohydrate dietary preferences.
- *Active Filter Feedback:* The "Low-Carb" pill button is visually highlighted, providing immediate confirmation of the active filtering state.
- *Targeted Results:* Showcases specific high-protein and low-carb options, such as "Turkey Lettuce Wraps," allowing for focused meal planning.
- *Data Consistency*: Filtered results maintain all essential metadata, including 3D icons, calorie counts, and easy-to-read difficulty levels.

## 4. Detailed Recipe Interface





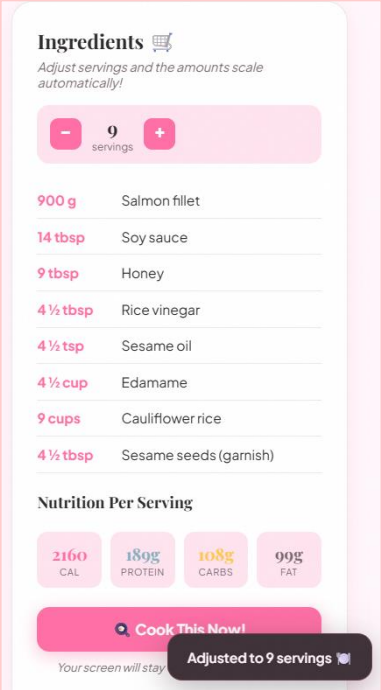

**Detailed Recipe Interface Description**

- *Immersive Hero Header:* The page features a sophisticated dark plum-colored header that displays the recipe title "Teriyaki Salmon Bowl" in bold typography, accompanied by a gourmet description and a high-quality 3D icon of the dish.

- *Quick Actions & Highlights:* Beneath the title, users find stylized badges for dietary tags (Healthy, Protein, Low-Carb) and essential metrics: a 35 min total time, 480 kcal count, and a Medium difficulty rating.
- *Interactive Controls:* Two prominent buttons, "Start Cook Mode" in vibrant pink and a outlined "Save Recipe" button, allow for immediate engagement with the content.

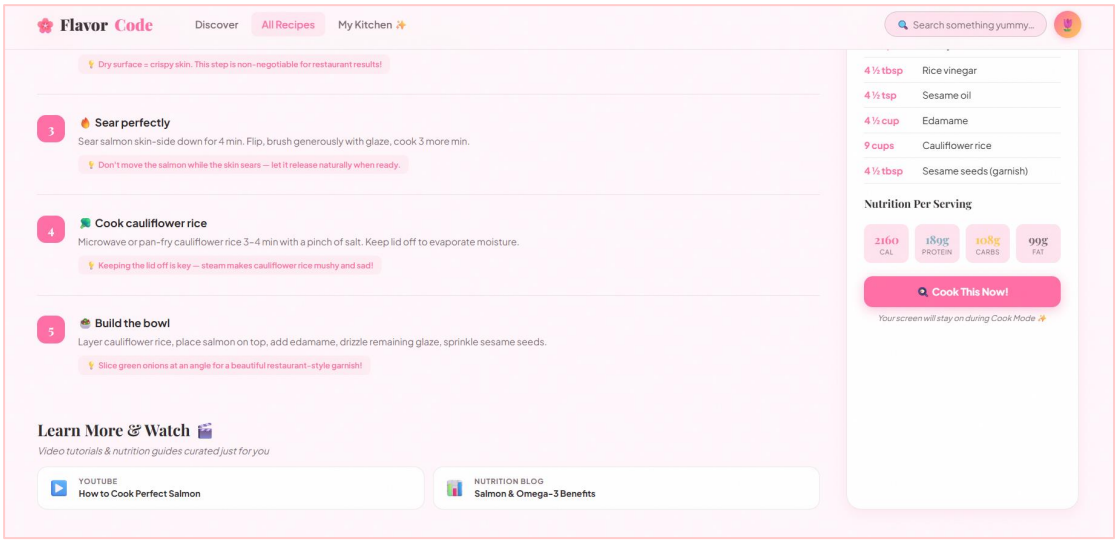
- *Structured "How to Make It" Section:* The core of the page offers a numbered, step-by-step guide (e.g., "Make the glaze," "Prep the salmon") with detailed instructions and "Pro Tips" highlighted in a soft pink callout for better cooking results.

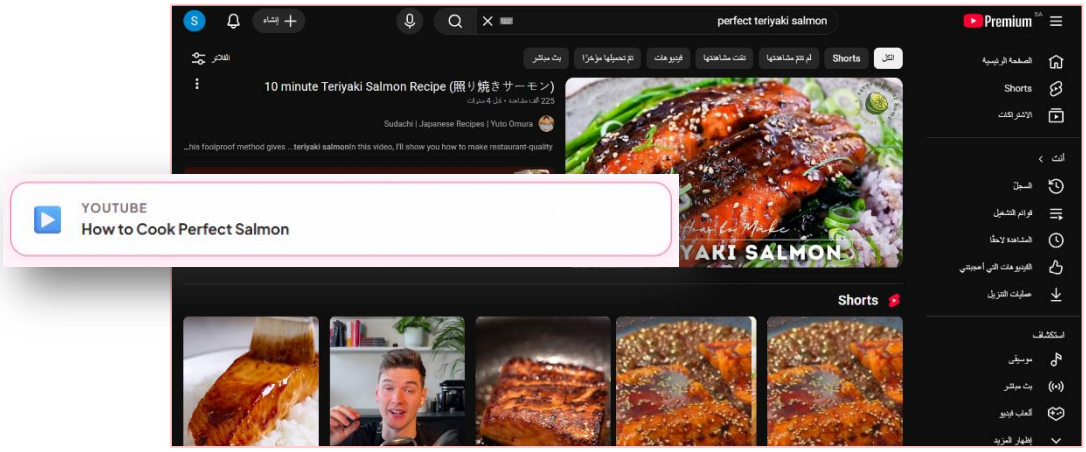
- *Dynamic Ingredients Sidebar:* A clean white sidebar on the right features an interactive serving adjuster that allows users to increase or decrease the number of servings. For example, the recipe was initially set to 2 servings with ingredient amounts such as “200 g Salmon fillet” and “3 tbsp Soy sauce.” After adjusting the serving size to 9 servings, all ingredient quantities automatically scaled in real time.

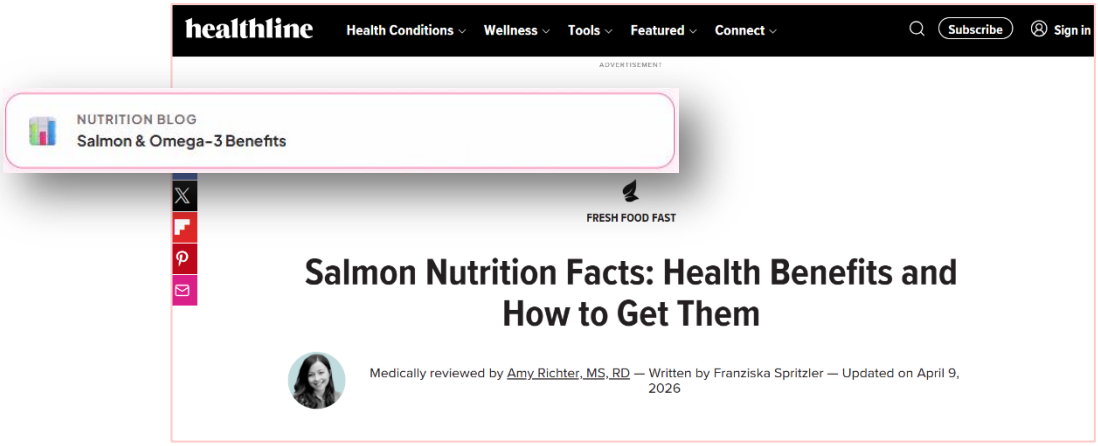



- *Comprehensive Nutritional Facts:* The sidebar also includes a “Nutrition Per Serving” section, providing a clear breakdown of calories, protein, carbs, and fat. The nutritional values automatically update when the serving size changes (e.g., from 480 calories at 2 servings to 2160 calories at 9 servings).

- *Educational "Learn More & Watch" Footer*: At the bottom, the interface integrates external resources, including a YouTube tutorial titled "How to Cook Perfect Salmon" and a nutrition blog post on "Salmon & Omega-3 Benefits" to enrich the user's knowledge.





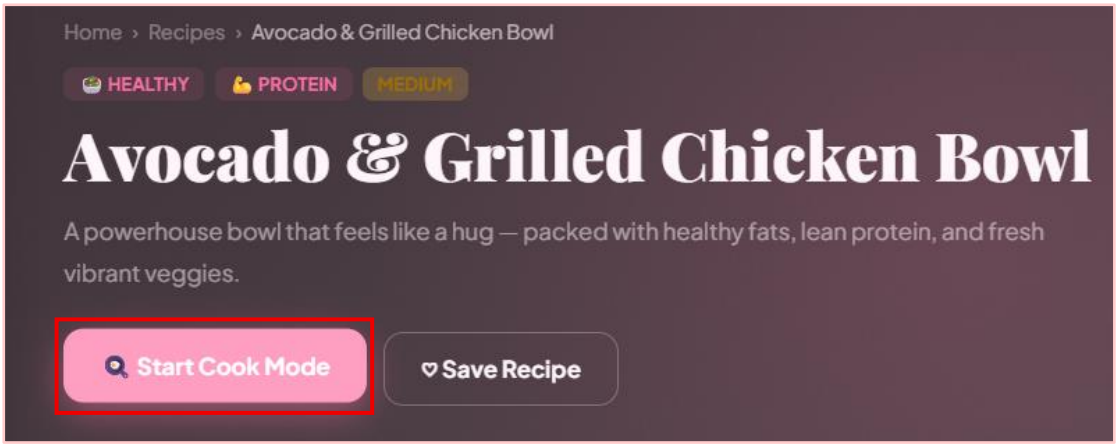

- *YouTube Tutorial Link:* A direct shortcut labeled "How to Cook Perfect Salmon" that redirects users to a curated video guide, ensuring visual support for technical cooking steps.

- *Interactive YouTube Overlay:* Upon clicking, the interface seamlessly bridges to a professional video tutorial (e.g., "10 minute Teriyaki Salmon Recipe") to enhance the learning experience.

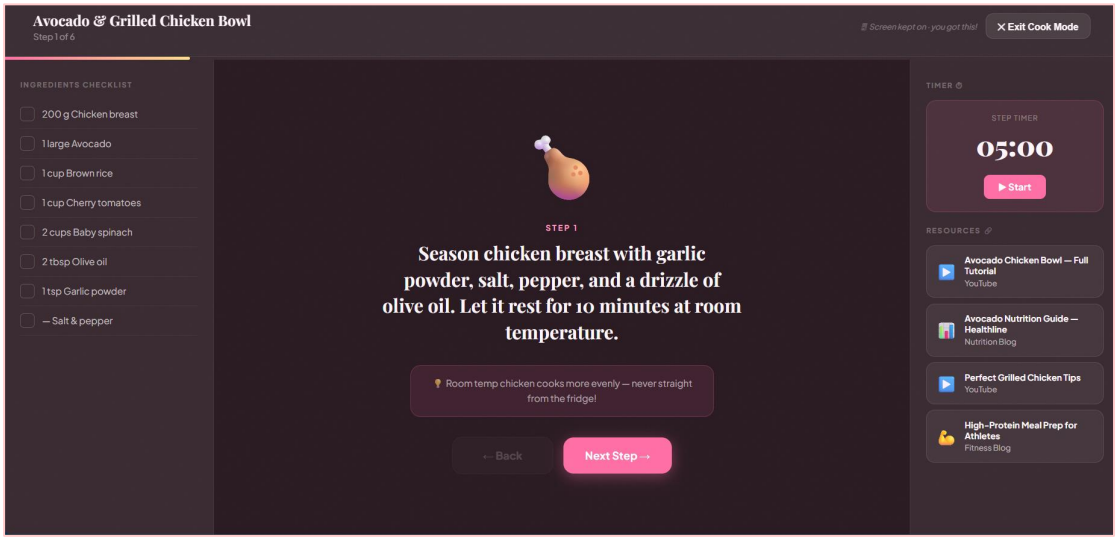
- *Nutrition Blog Access:* A dedicated link titled "Salmon & Omega-3 Benefits" that connects users to high-quality health articles, such as those from Healthline.

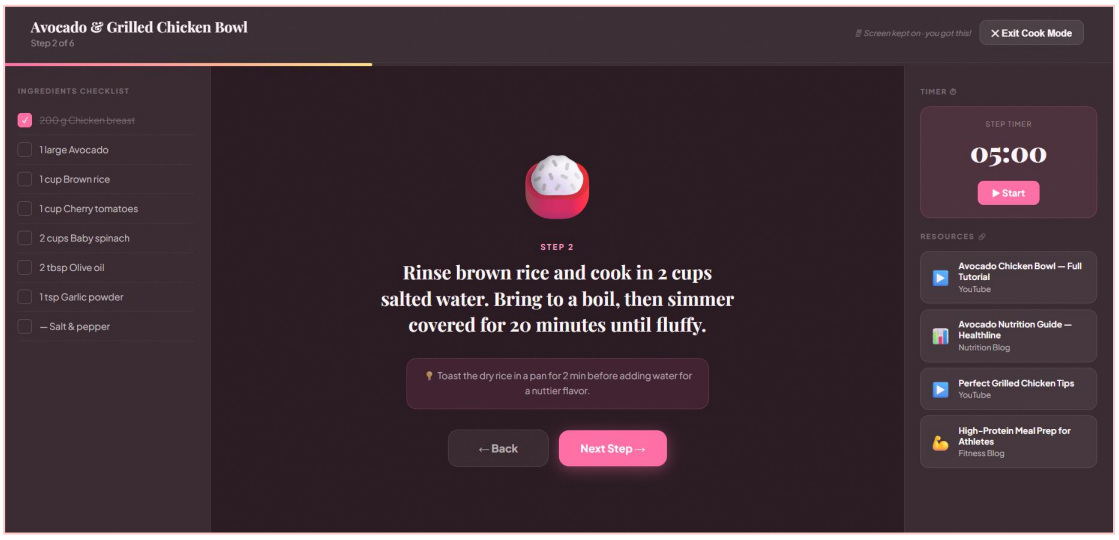
- *Science-Backed Content:* The redirected blog provides medically reviewed nutrition facts, covering topics like heart health, brain health, and inflammation.

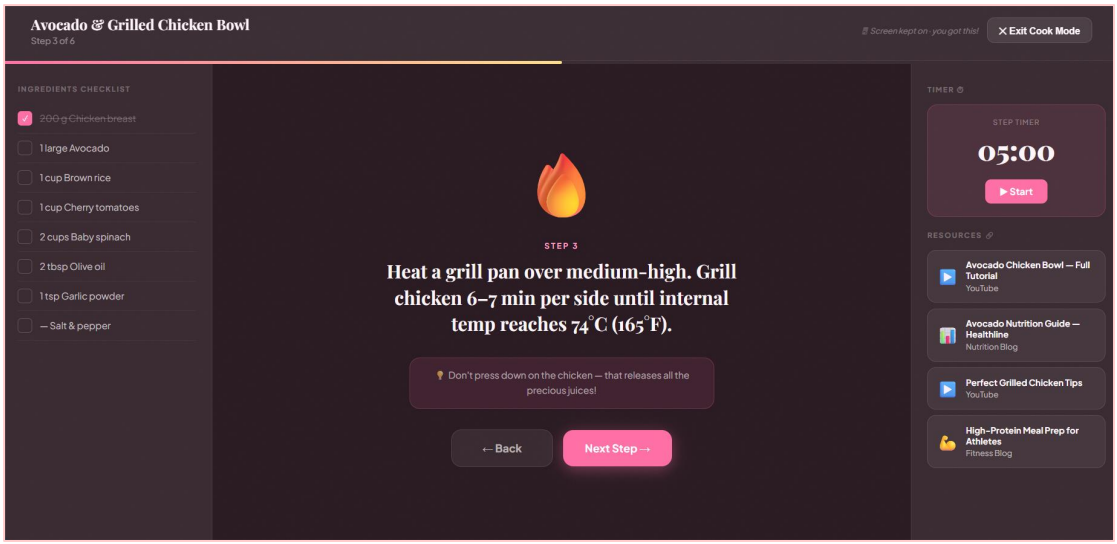
## 5. Cook Mode Interface

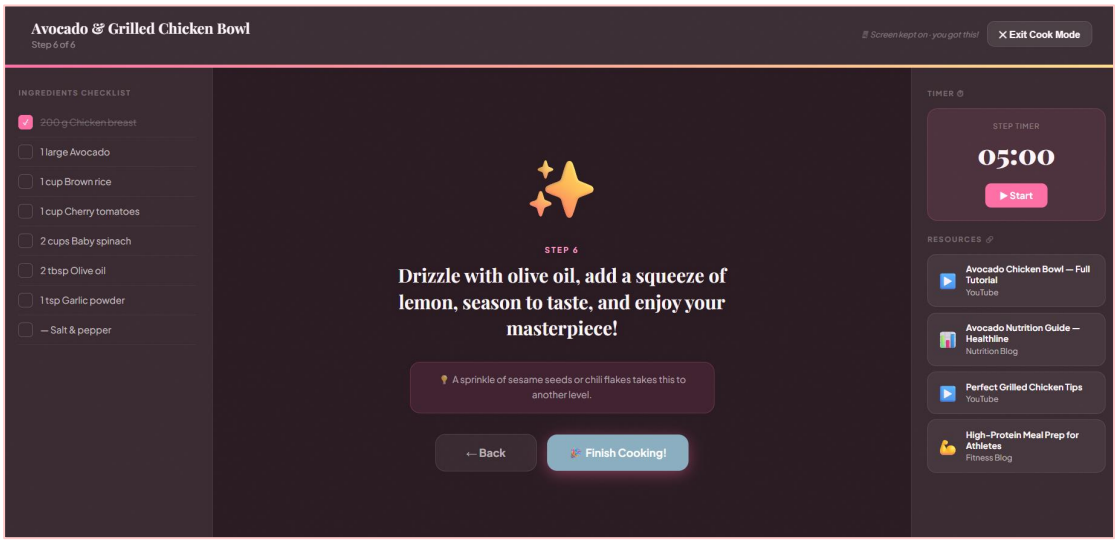



Initiation of Interactive Guidance Clicking the "Start Cook Mode" button transitions the user from a static recipe view to a dynamic, full-screen cooking assistant.





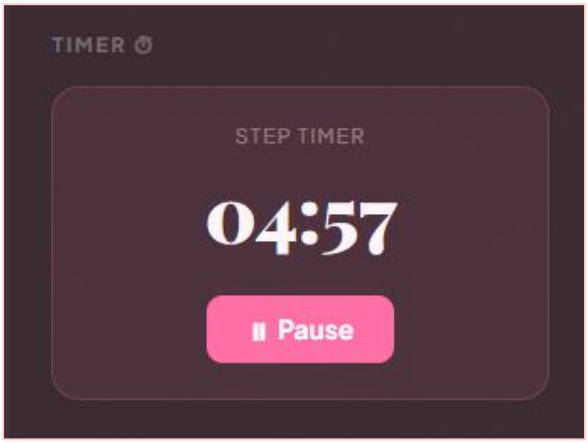



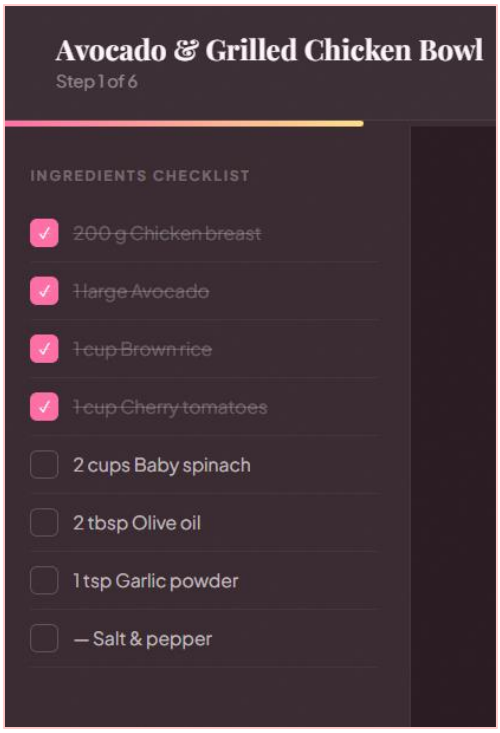

Flexible Step Navigation.

The interface provides a "Back" button, allowing users to easily revisit previous instructions and correct any mistakes during the preparation process.

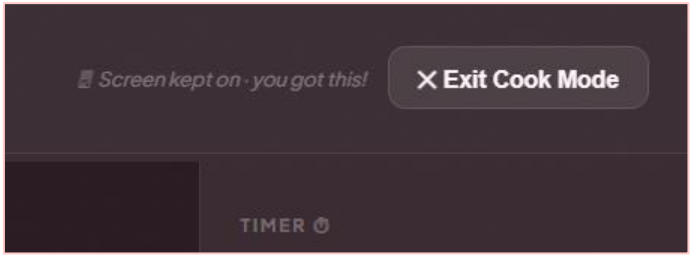
## Cook Mood Interface Description

- Integrated Step Timer A dedicated Step Timer is included to help users manage time-sensitive tasks with precision, featuring "Start" and "Pause"functionality.



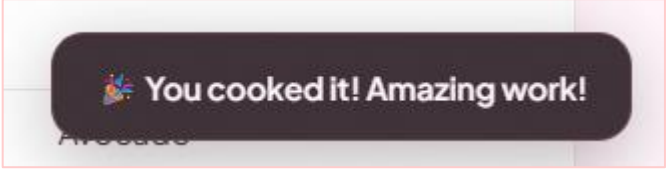



- Interactive Ingredients Checklist A sidebar checklist allows users to mark off ingredients
  as they are used, ensuring no component is missed during complex assembly.

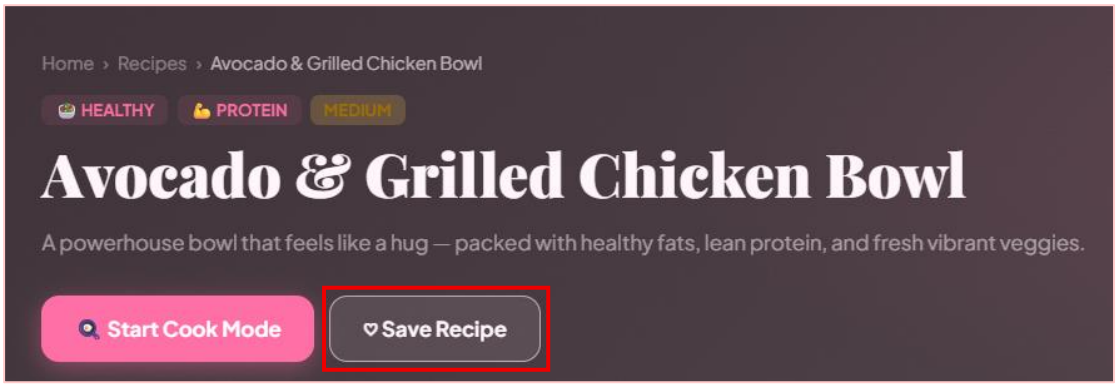



- Control & Exit Functionality Users have the freedom to leave the interactive session at any time via the "Exit Cook Mode" button, providing a non-restrictive user experience.

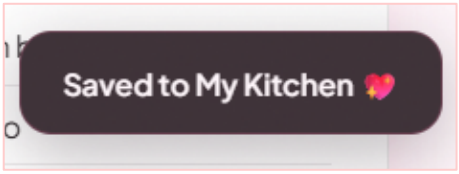
- Post-Cooking Feedback Upon finishing the recipe, a celebratory Feedback notification ("You cooked it! Amazing work!") appears, rewarding the user and providing emotional fulfillment.



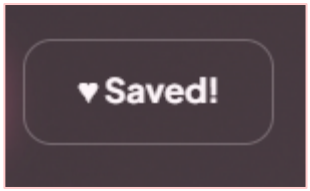
## 6. Recipe Saving & Personalization Features



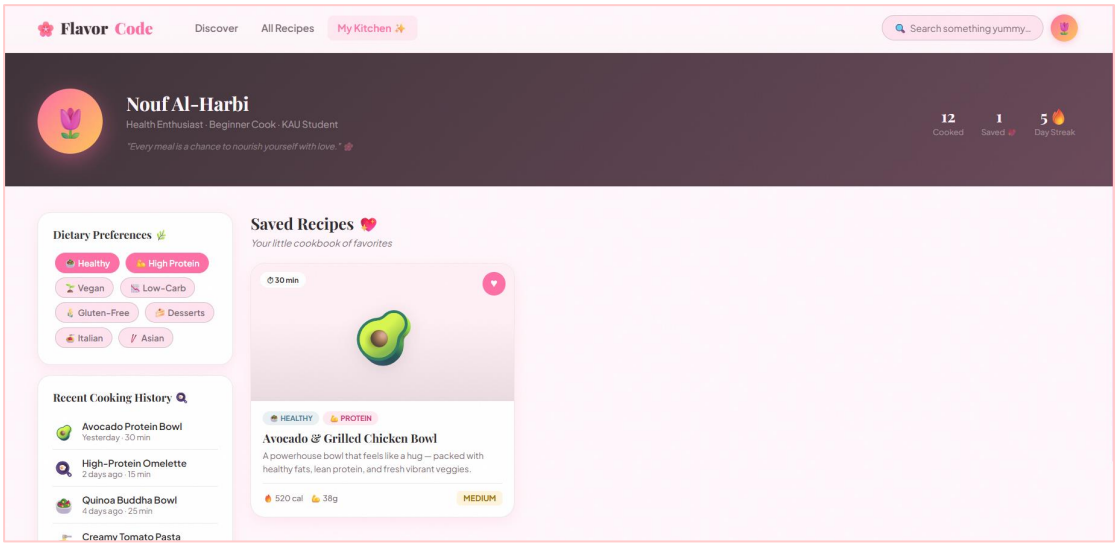

- Save Recipe Action | The interface features a clear, outlined "Save Recipe" button with a heart icon, allowing users to bookmark their favorite dishes for future access.




- Immediate User Feedback | Upon clicking the save button, a small dark toast notification appears with the message "Saved to My Kitchen", providing instant confirmation that the action was successful.



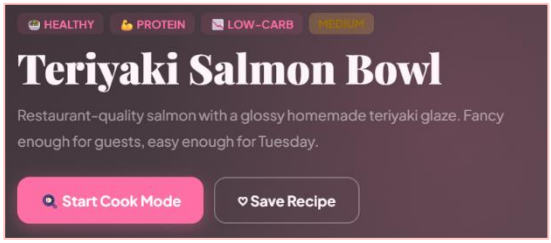

## 7. My Kitchen Interface



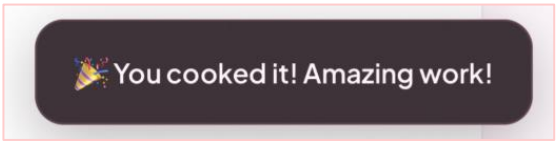
- *User Header:* The top of the page displays the user's name and greeting to confirm the personal account state.
- *Quick Stats Summary:* A data section shows the user's current kitchen activity, including the total number of "Saved Recipes" and "Cooked Recipes".
- *Saved Recipes Section:* A dedicated gallery area labeled "Saved Recipes" that houses all the dish cards previously bookmarked by the user.
- *Dietary Preferences Settings:* A configuration area where users can select and lock their specific dietary needs (e.g., Vegan, Gluten-Free, High Protein) to personalize their discovery feed.
- *Favorite Categories Toggle:* A selection of category buttons that allows the user to quickly filter their own saved collection by meal type or prep time.
- *Interactive Recipe Cards:* The page contains functional cards for each saved dish, providing direct buttons to either "Start Cook Mode" or "Remove from Kitchen".
- *Navigation Persistence:* The standard top navigation bar remains available, with the "My Kitchen" tab highlighted to indicate the current active view.

## 3.2 Design Principles Applied in Flavor Code

- ***Visibility***: The interface ensures that the most important actions are impossible to miss by using high-contrast colors and clear icons. For example, the "Start Cook Mode" button is highlighted in a vibrant pink against a dark background to draw immediate attention.



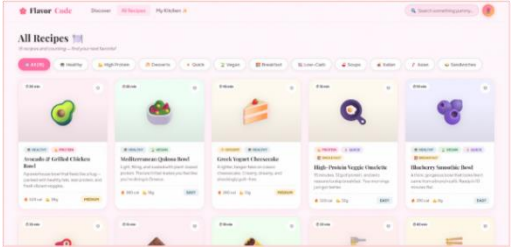

- ***Feedback***: The system provides instant confirmation for every user action t o keep the user informed. For example, when a user clicks 'Save Recipe,' a toast notification appears stating 'Saved to My Kitchen.' Furthermore, upon completing a recipe in Cook Mode, the system rewards the user with a celebratory message: 'You cooked it! Amazing work!', confirming the successful completion of the task.





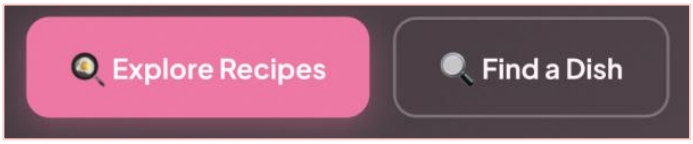


- ***Consistency***: Flavor Code maintains a uniform visual language across all pages. All recipe cards follow the same structural layout—including the placement of 3D icons, calorie counts, and prep times—making the interface predictable.

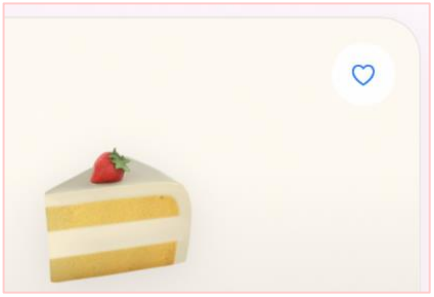


- ***Affordance***: The design utilizes visual cues and physical attributes that suggest how users should interact with the site:

    1. *Rounded Corners (Border Radius):* Most interactive elements, such as the search bar and recipe cards, feature rounded corners. This "softness" makes the elements appear as distinct, clickable objects rather than flat text, inviting users to engage with them.




    2. *Iconography*: Standardized icons, such as the magnifying glass for search and the heart for saving, act as strong affordances. They bridge the gap between form and function, allowing users to understand the purpose of a button immediately without reading the label.



- ***Constraints***: To prevent user errors, the app uses graphical constraints like horizontal category pills. These limit the initial view to specific dietary needs (e.g., Vegan), preventing the user from feeling overwhelmed.



## 3.3 Cognitive Processes Supported by the Design

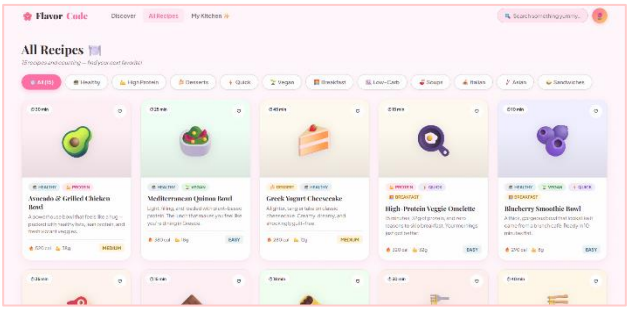
- ***Attention***: The "Discover" page manages the user's attention through a visual hierarchy. Large headlines like "Cook Cuter, Eat Better" and the "Popular Right Now" section use size and bold typography to guide the user's eyes toward trending content first.



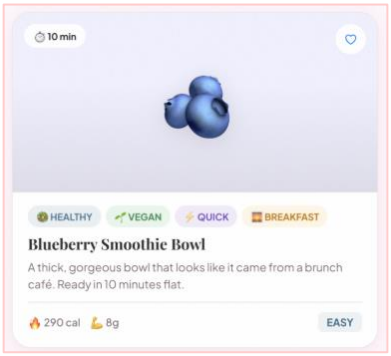




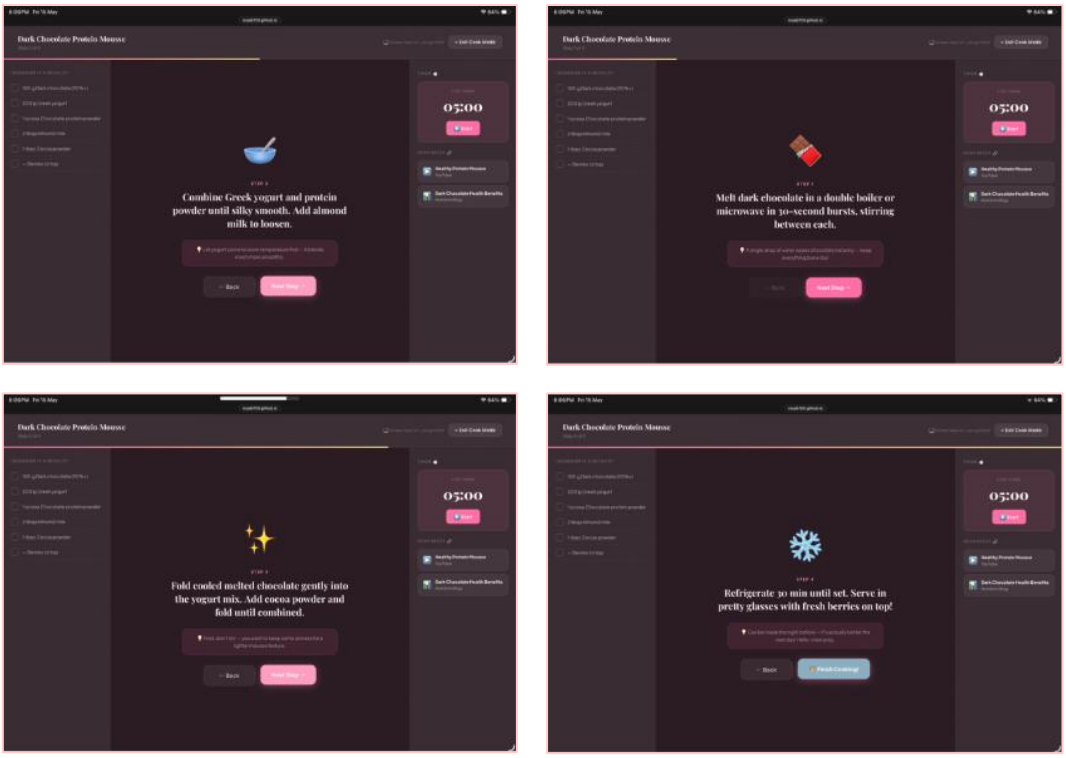
- ***Perception***: The app enhances perception by using 3D iconography and ample white space. This allows users to quickly identify different food types (like a steak icon for meat or a cake for dessert) without needing to read extensive text.



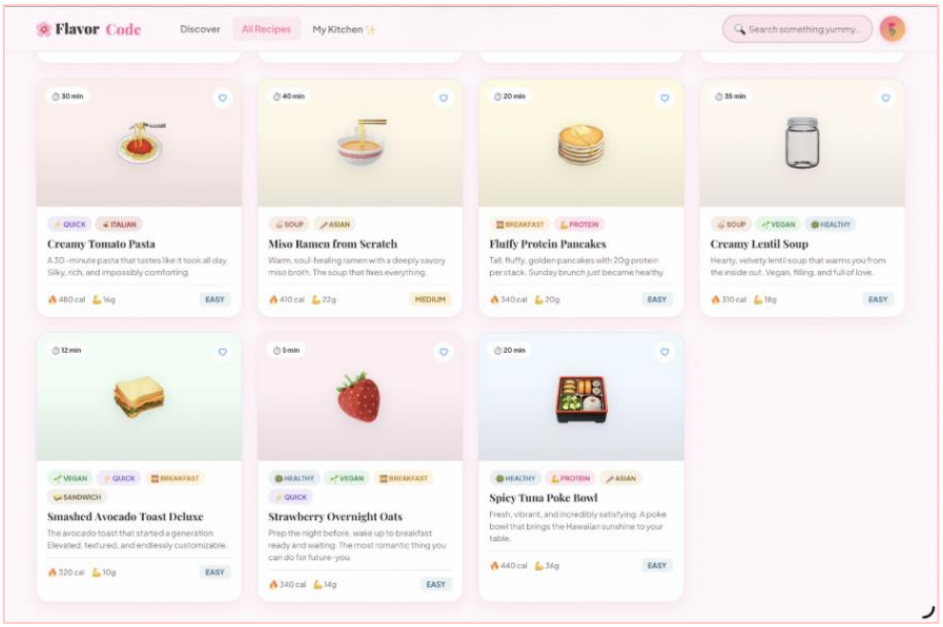

- ***Memory***: To reduce the cognitive load on human memory, the app displays all essential metadata (calories, time, difficulty) directly on the cards. This ensures users do not have to memorize details while browsing through the gallery.



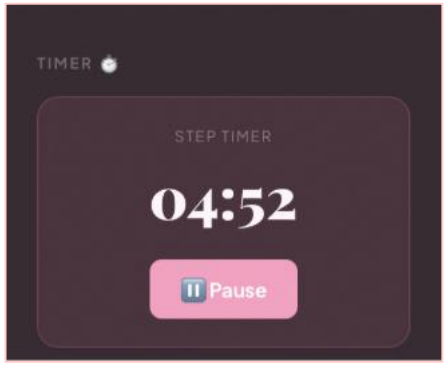

- ***Learning***: The "Cook Mode" supports learning through doing. By breaking down complex recipes into single, numbered steps with a progressive "Next Step" button, the app guides users through the process and builds their confidence.

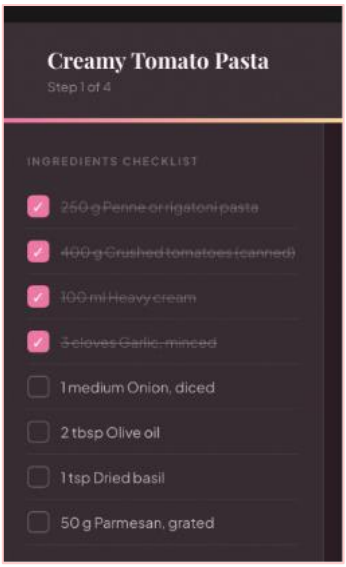



- ***Decision-Making***: The app facilitates fast and informed decisions by displaying at-a-glance metadata. Users can choose a meal that fits their health goals and time availability immediately by looking at the tags on each card.

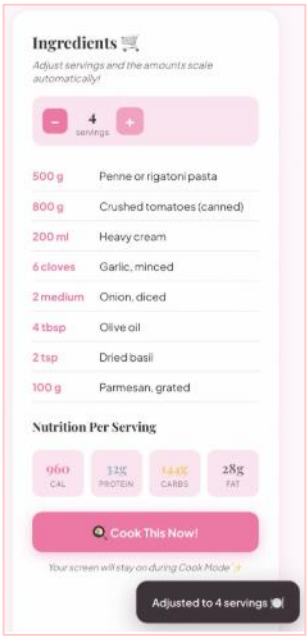


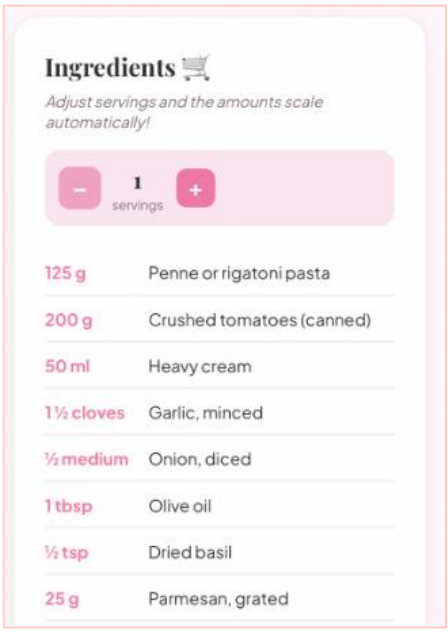
- ***Problem-Solving:*** Within Cook Mode, the app helps users solve the problem of multi-tasking. The Step Timer and Ingredients Checklist provide a structured environment that prevents the user from losing track of time or steps.




- ***Reasoning:*** The Dynamic Serving Adjuster supports logical reasoning. When a user changes the number of servings, the system automatically recalculates ingredient amounts, allowing the user to understand their prep needs without manual math.



# Part 4 – Evaluation

## 4.1 Tasks Used for Evaluation

Participants were asked to complete the following tasks using the Cook website prototype:

- Sign Up / Log In.
- Browse the homepage and explore recipe categories.
- Search for a recipe and view its details.
- Browse Recipes by Categories and Dietary Filters.
- Save a Favorite Recipe.
- Use Step-by-Step Cook Mode.
- Use the Cooking Timer Feature.
- Navigate between pages and return to the homepage.

## 4.2 Participants

The prototype was evaluated by 31 participants. The evaluation was conducted with a group of participants aged between 18 and 45 years old, representing the target users of the Flavor Code application. Most participants were university students and young adults who regularly use mobile applications and search for cooking recipes online. Participants had different levels of cooking experience, ranging from beginners to experienced home cooks.

## 4.3 Testing Environment

The prototype testing was conducted in a comfortable and controlled environment using laptops and mobile devices. Each participant was asked to complete a set of realistic cooking-related tasks while interacting with the Flavor Code prototype. Participants were encouraged to freely explore the application features, including recipe browsing, Cook Mode, timers, and saving recipes. Observations and feedback were collected during the testing sessions to evaluate usability, navigation clarity, and overall user satisfaction.

## 4.4 Usability Goals and User Experience Goals

The evaluation results showed that the Cook website successfully achieved important usability and user experience goals.

### 4.4.1 Usability Goals

The results indicate that users could complete tasks efficiently with minimal confusion or errors.

- **Effectiveness:** Most participants successfully completed the required tasks such as recipe searching and page navigation without major problems.

  93.5% strongly agreed that navigation between pages was clear and simple.

- **Efficiency:** Users were able to find recipes quickly and interact with the website smoothly.

  83.9 strongly agreed that the website was easy to use.

  90.3% strongly agreed that they were able to find recipes quickly.

- **Ease of Use:** The majority of participants strongly agreed that the website interface was simple and user-friendly.

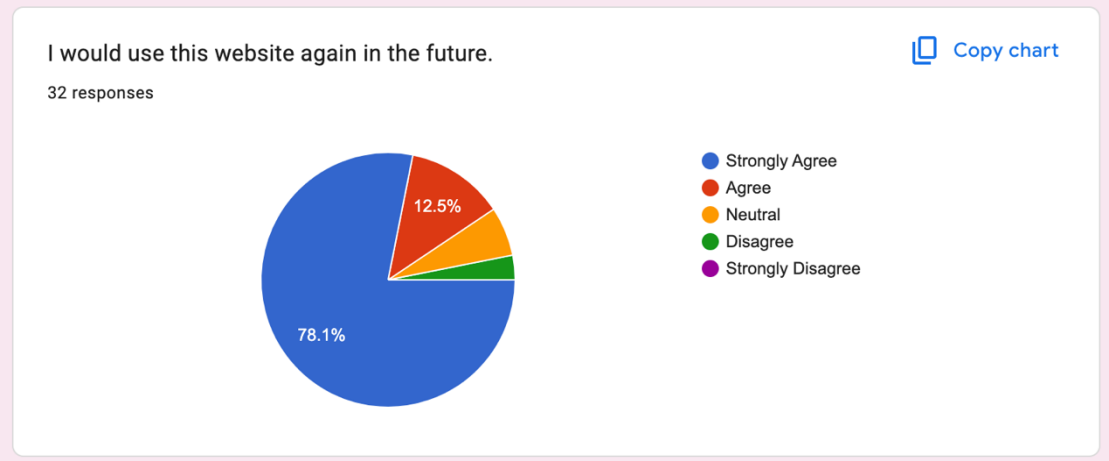
  93.5% strongly agreed that the interface design was visually attractive.

### 4.4.2 User Experience Goals

The website also provided a positive user experience, Participants particularly liked the clean design, easy navigation, and organized recipe categories.

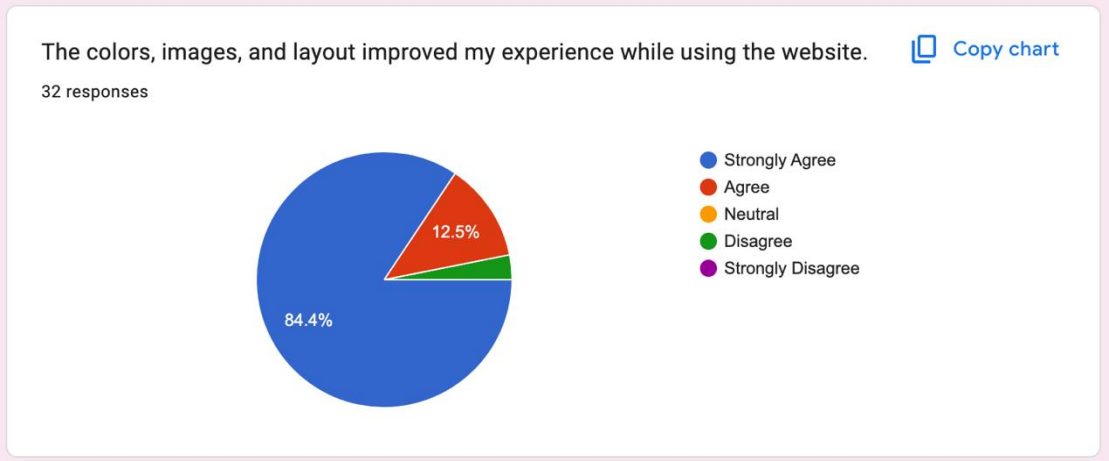
- **Satisfaction:** Most users expressed high satisfaction with the Cook website and indicated that they would use it again in the future.

    77.4% strongly agreed that they would use the website again in the future.



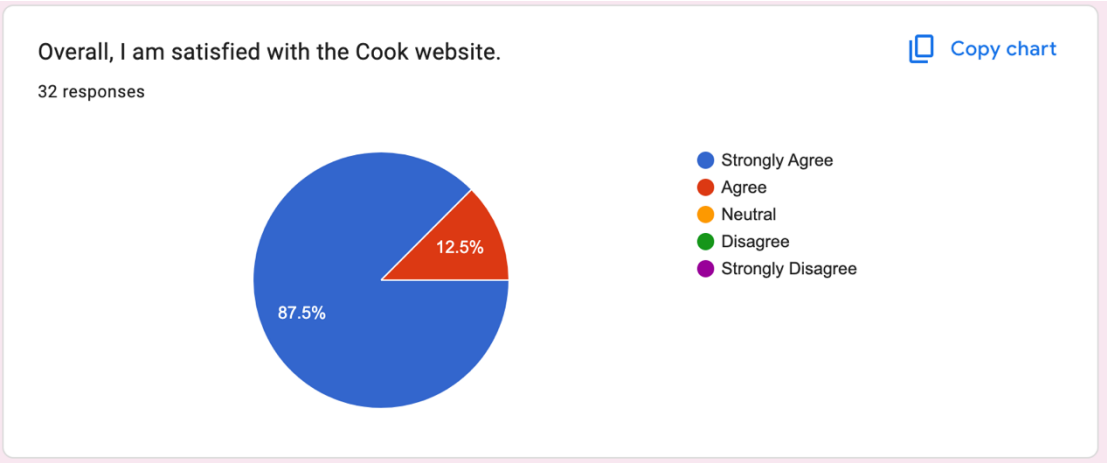

- **Aesthetic Appeal:** Participants liked the visual design, colors, layout, and recipe organization.

    83.9% strongly agreed that the colors, images, and layout improved their experience.

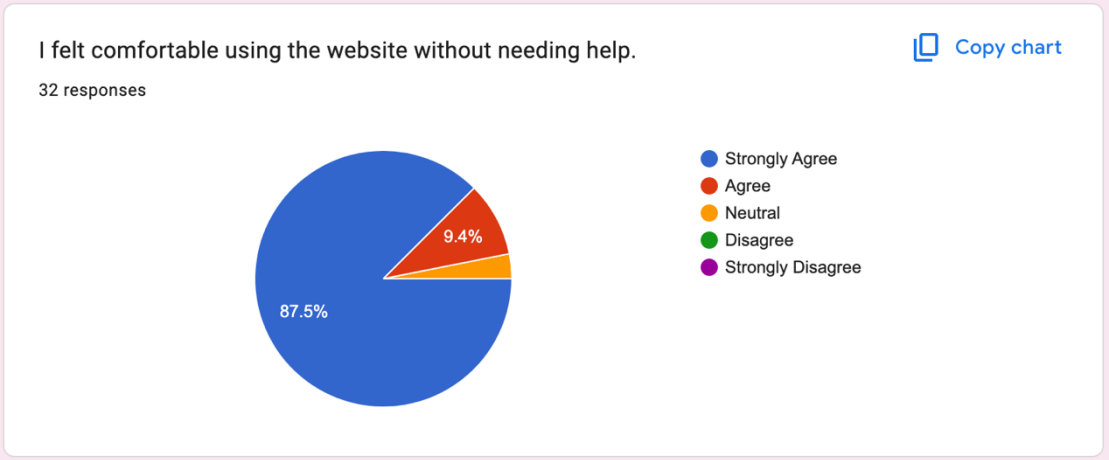


- **Enjoyment:** Users reported a comfortable and enjoyable experience while browsing the website and exploring recipes.

87.1% strongly agreed that they were satisfied overall with the Cook website.




87.1% strongly agreed that they felt comfortable using the website without needing help.



## 4.5 Result Analysis

Most users completed the required tasks successfully and reported a high level of satisfaction with the interface. The charts show that users found the website visually appealing, easy to navigate, and efficient for finding recipes.

The usability evaluation focused on several important tasks, including logging into the website, searching for recipes, navigating between pages, and saving favorite recipes. Participants' feedback was analyzed based on usability, efficiency, ease of navigation, and overall satisfaction.

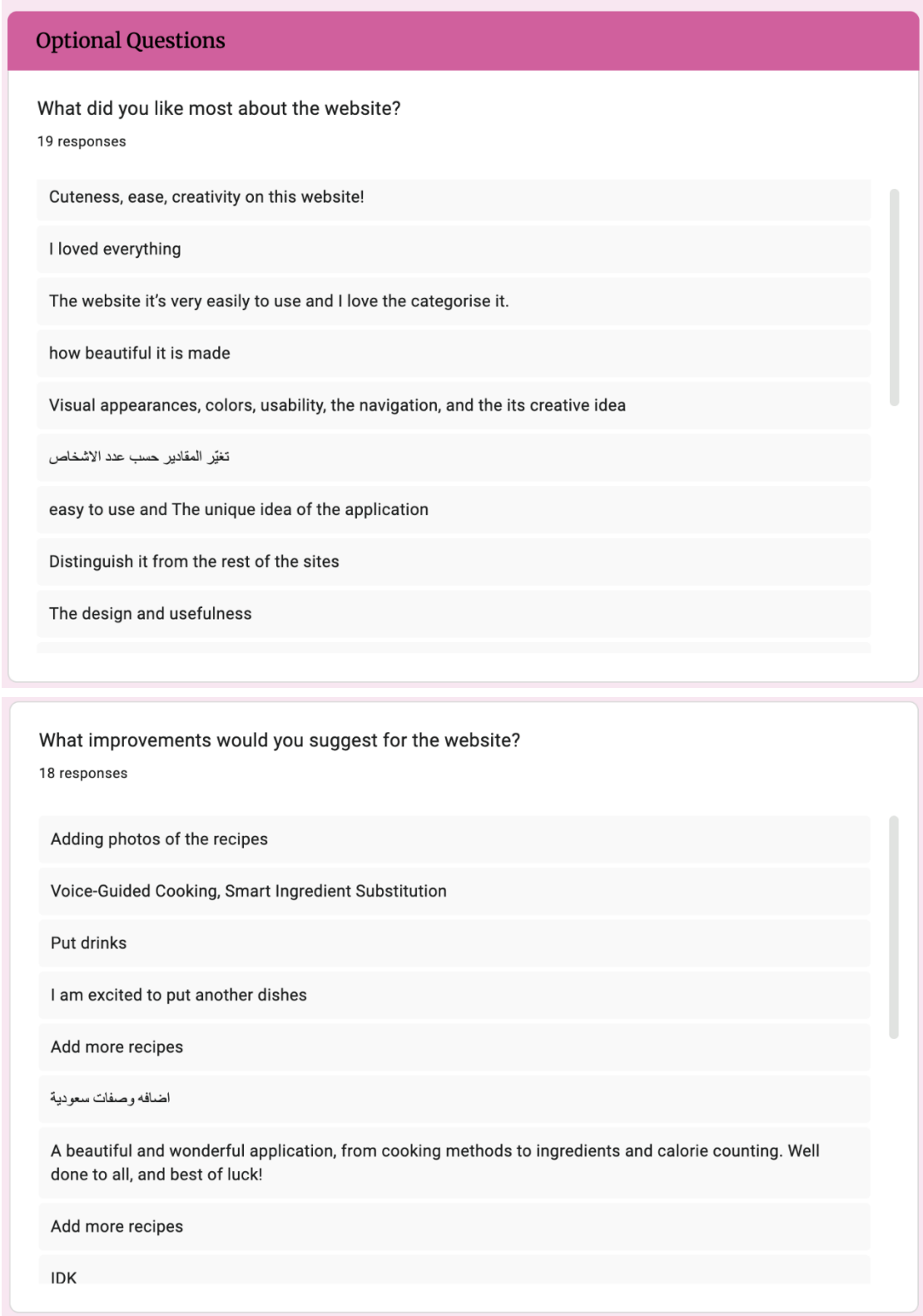
- **Log In / Sign Up:**
  Most participants completed the login and sign-up process successfully and without confusion. Users found the authentication process simple and clear, which helped them access personalized features such as saved recipes and profile settings quickly.

- **Recipe Search and Navigation:**
  Participants were able to search for recipes efficiently and move between pages smoothly. Around 90.3% of participants strongly agreed that they could find recipes quickly, while 93.5% strongly agreed that navigation between pages was clear and simple. These results indicate that the website structure and organization were effective and user-friendly.

- **Visual Design and User Satisfaction:**
  The interface received highly positive feedback regarding colors, layout, and overall appearance. About 93.5% of participants strongly agreed that the interface design was visually attractive, and 87.1% expressed overall satisfaction with the Cook website. Participants particularly appreciated the clean design and organized recipe categories.

- **Comfort and Ease of Use:**
  Most users reported feeling comfortable while using the website without needing assistance. The results showed that the website provided a smooth and enjoyable user experience with minimal usability issues.

Some participants suggested future improvements, including:



**Optional Questions**

What did you like most about the website?

19 responses

- Cuteness, ease, creativity on this website!
- I loved everything
- The website it's very easily to use and I love the categorise it.
- how beautiful it is made
- Visual appearances, colors, usability, the navigation, and the its creative idea
- تغيّر المقادير حسب عدد الاشخاص
- easy to use and The unique idea of the application
- Distinguish it from the rest of the sites
- The design and usefulness

What improvements would you suggest for the website?

18 responses

- Adding photos of the recipes
- Voice-Guided Cooking, Smart Ingredient Substitution
- Put drinks
- I am excited to put another dishes
- Add more recipes
- اضافه وصفات سعودية
- A beautiful and wonderful application, from cooking methods to ingredients and calorie counting. Well done to all, and best of luck!
- Add more recipes
- IDK

Overall, the evaluation results demonstrate that the Cook website successfully achieved its usability and user experience goals by providing an attractive, efficient, and easy-to-use cooking platform.

# References

Nielsen, J. (1994). *Usability Engineering. Morgan Kaufmann.*

Norman, D. (2013). *The Design of Everyday Things. Basic Books.*

Preece, J., Rogers, Y., & Sharp, H. (2019). *Interaction Design: Beyond Human-Computer Interaction.* Wiley.

# Appendix

1. First Questionnaire:
   https://docs.google.com/forms/d/e/1FAIpQLSdQtl8Nx9eO2uzs8vKkvkKkY7vaLyYfvNpRGuzJjzdOpcmyoQ/viewform

2. Second Questionnaire:
   https://docs.google.com/forms/d/e/1FAIpQLSe5wEzhySudZ6bs63b90aP0bCThI7XH6oCmbcyZAeUodYQ7Cg/viewform

3. Prototype:
   https://flavourcode.netlify.app/